

www.vitrox.com

V810i Controller Subsystem





V810i Controller Subsystem

Automated Board Inspection

This training material is applicable to below models:

Machine Model	
V810i STD	☑
V810i XXL	☑
V810i S2EX	☑
V810i XLT	☑
V810i XLTM	☑
V810i S3	☑

Revision History

Rev	Description
3	include machine XLTM & S3 include Atom PC (for Vicat IO, Panasonic drive) include S2Rev5 server include white Cisco switch include network card



Hardware Overview

Automated Board Inspection

Main Controller - Main PC of V810i system to process and analysis data.

Current V810 machine server equipped with Windows 10 Enterprise LTSC 21H2.

Vitrox offer optional upgrade solution for previous server equipped with Windows 8 or Windows 10 Pro.

Remote Motion Server - An interface as communication device between SSCA/MCA Box and Main controller via MEI card.

Installed RMS Driver, Clear Path Driver and Oriental motor MEXE02 software driver.

Remark: additional Kollmorgen WorkBench software driver if machine equip with AKD Kollmorgen Drive

Cisco Switch - an interface as communication device for main controller, X-Ray Cameras, RMS for data transferring purpose.

	User Name	Password	IP Address
Main Controller	Administrator	Please!	192.168.128.60
RMS	Administrator	Please!	192.168.128.70
Cisco Switch (black)	cisco	cisco	192.168.128.122
Cisco Switch (white)	admin	cisco	192.168.128.122



V810i Series 1

- 4 Image Reconstruction Processors
- Dell R720 Master Controller
- Dell Gigabyte Switch
- Capacity Window 7 with 64 bits



V810i Series 2

- Scan Path Merging (SPAM)
- Real time Predictive Slice Height
- Single Unified Management Office (High speed processor)
- 128GB RAM Capacity - Basic package
- Upgrade-able to 256GB RAM - with PO
- 64 bits Image Reconstruction Processor



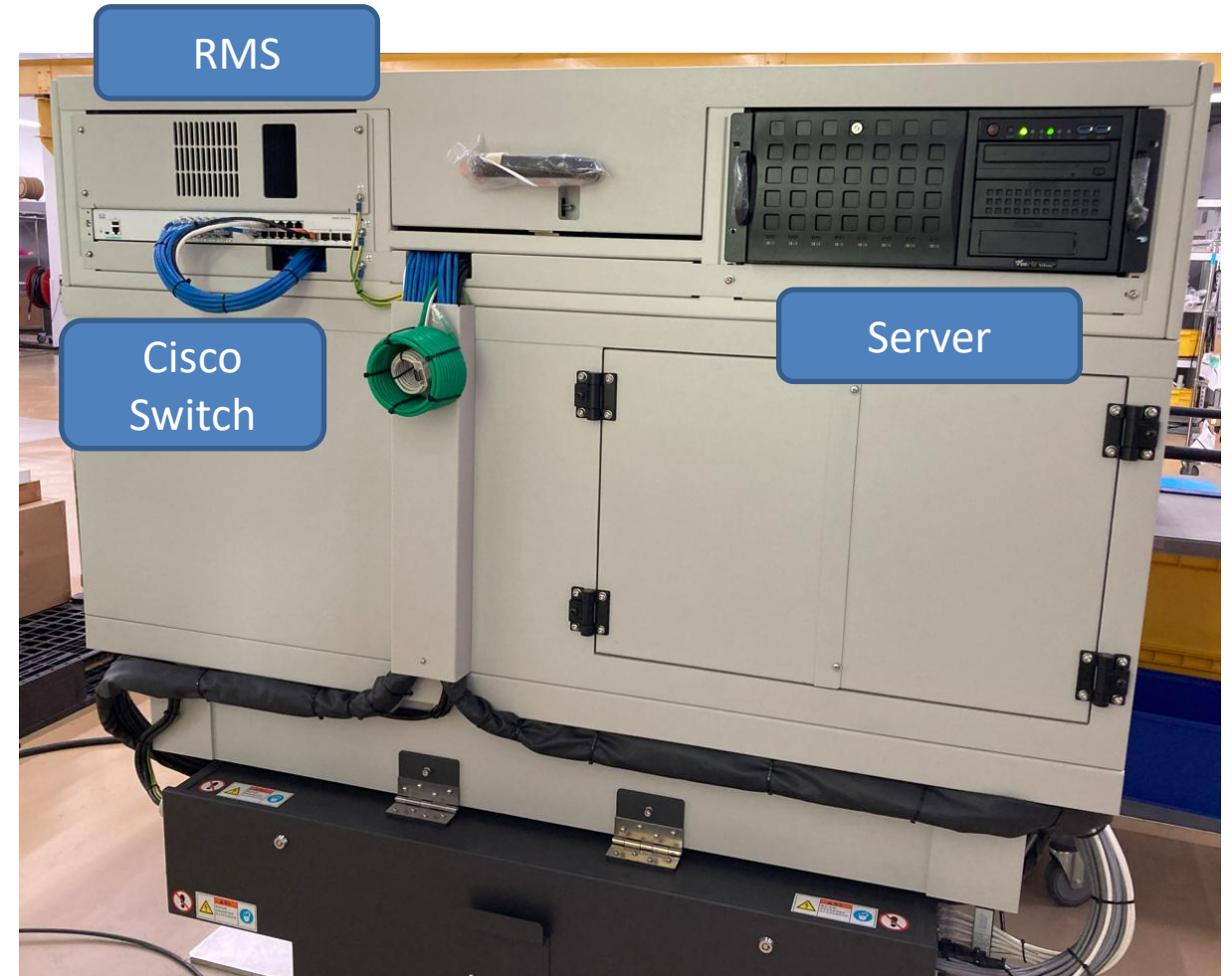
V810i Series 1



V810i Series 2

ABI Hardware Overview (S2EX)

Automated Board Inspection



Autom

RMS

Cisco
Switch

Server

29/11/2018 16:46

Cisco
Switch

Server

Automated Board Inspection



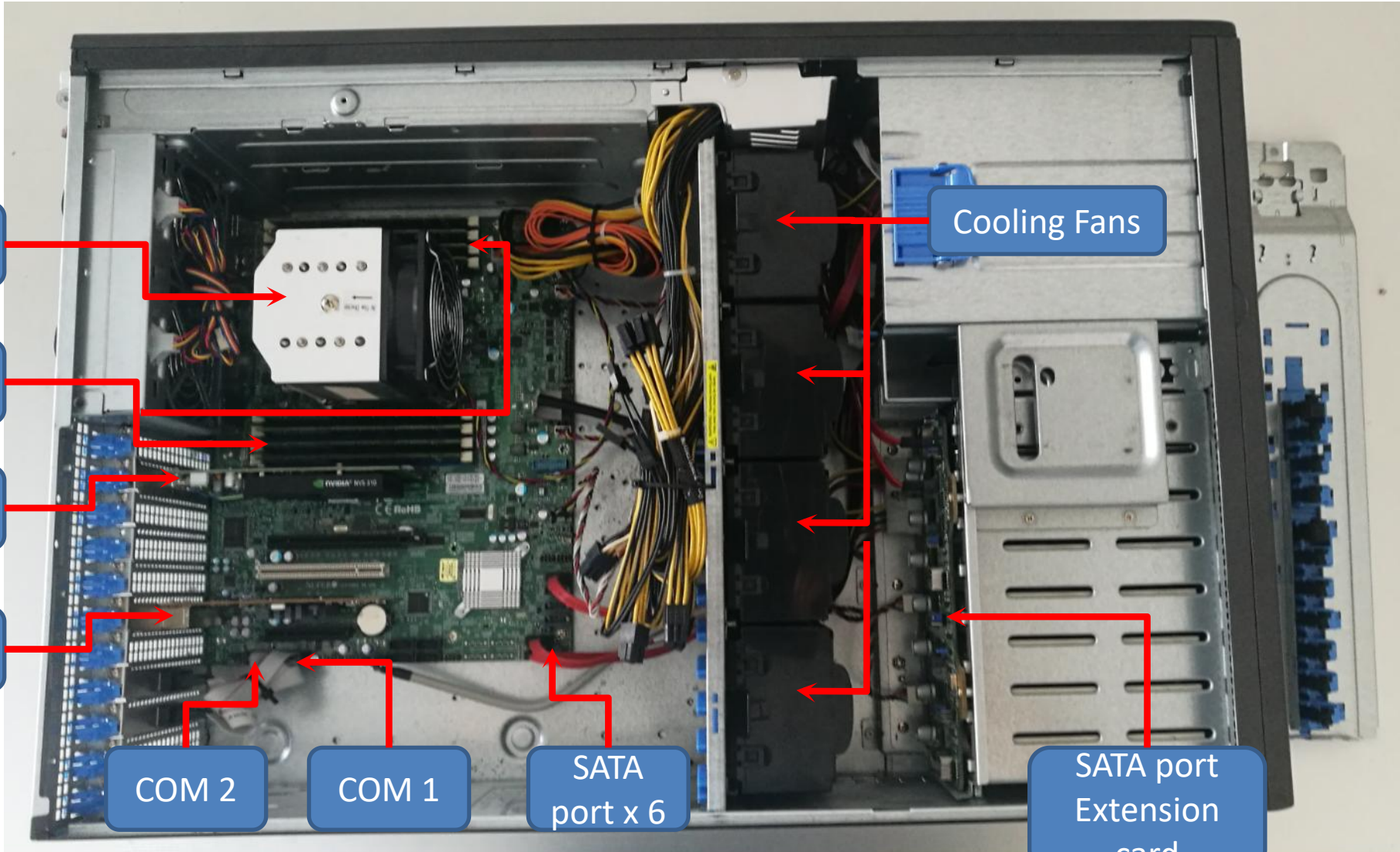
S2Rev2

Item	Descriptions
Motherboard	Supermicro X9SRA
Processor	Intel Xeon E5-2687W
Memory	128GB (8x 16GB DDR3 1600MHz ECC/Reg server memory)
Hard Drive	2TB - 2x WD 1TB 7200RPM 128MB CACHE 3.5IN SATA Enterprise Class HDD
Optical Drive	24X DVD RW Optical Drive
Display Adapter	NVIDIA Quadro NVS 310 DVI
PSU	1400W redundant PSU



S2Rev3

Item	Descriptions
Motherboard	Supermicro X9SRA
Processor	Intel Xeon E5-2687W
Memory	128GB (8x 16GB DDR3 1600MHz ECC/Reg server memory)
Hard Drive	2TB - 2x WD 1TB 7200RPM 128MB CACHE 3.5IN SATA Enterprise Class HDD
Optical Drive	24X DVD RW Optical Drive
Display Adapter	NVIDIA Quadro NVS 310 DVI
PSU	920W redundant PSU



Processor with
Cooling fans

DDR 3 RAM

Graphic Card
(GPU)

Network Card

COM 2

COM 1

SATA
port x 6

Cooling Fans

SATA port
Extension
card



Main Controller Overview

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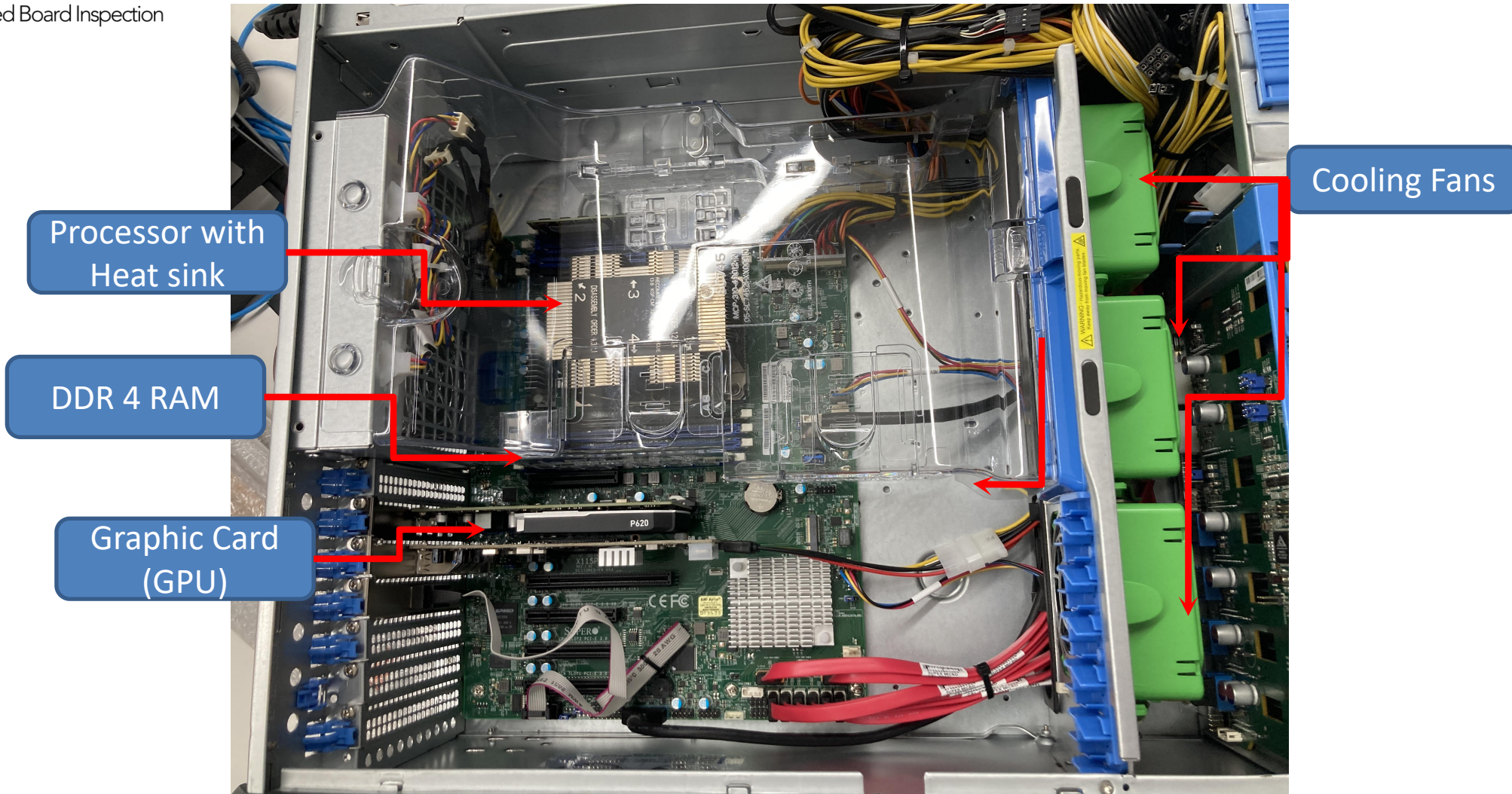
S2Rev4/
S2Rev5/
S2Rev5.1

2160-0129

S2Rev5.2
2160-0185

Item	Descriptions
Motherboard	Supermicro X11SPL-F
Processor	Intel Xeon Gold 6134
Memory	128GB (8x16GB / 2x 64GB DDR4 2666MHz ECC server memory)
Hard Drive	2TB - 2x WD 1TB 7200RPM 128MB CACHE 3.5IN SATA Enterprise Class HDD
Optical Drive	24X DVD RW Optical Drive
Display Adapter	NVIDIA Quadro P620 2GB
PSU	920W redundant PSU

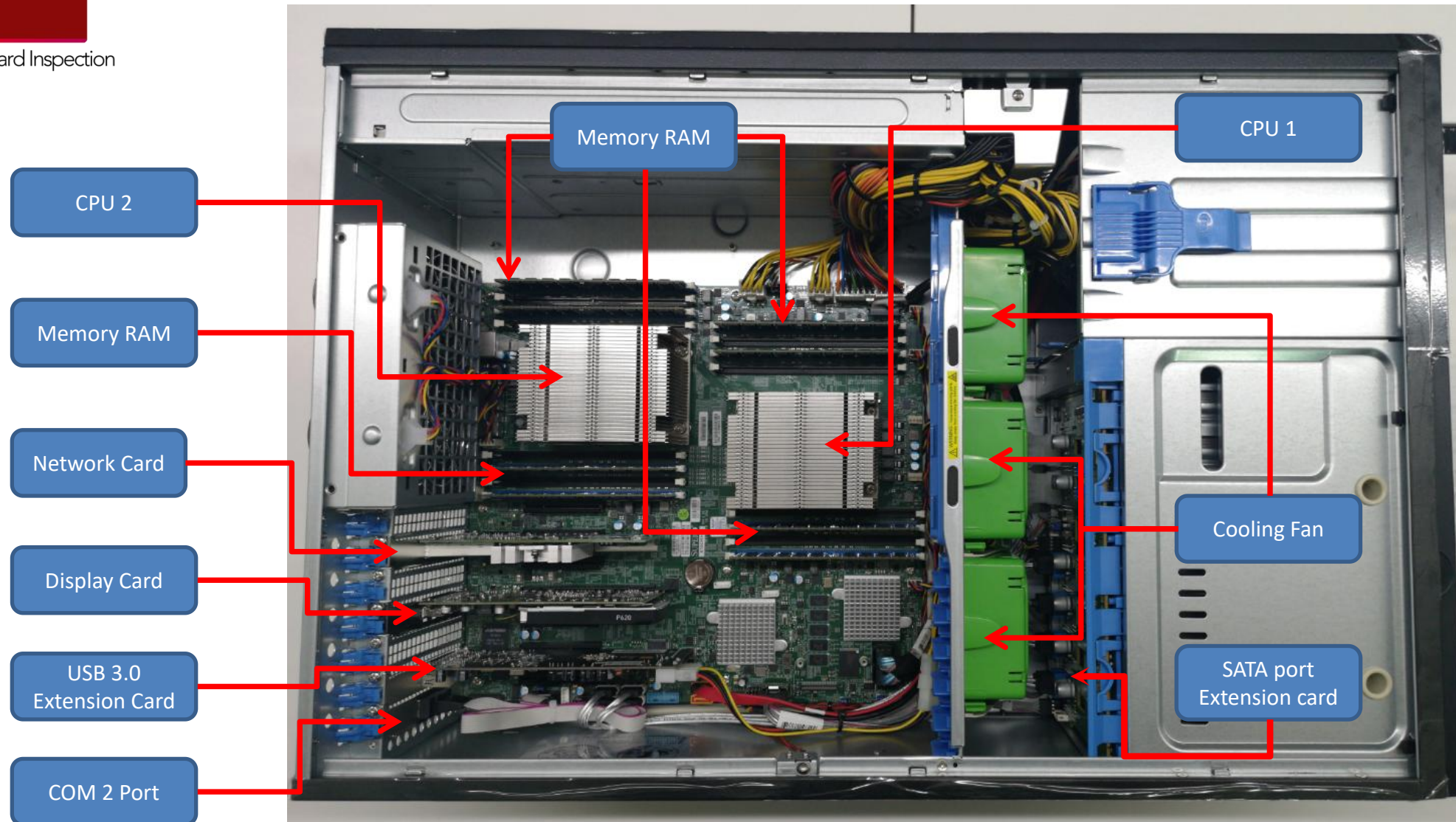
Item	Descriptions
Motherboard	Supermicro X11SPL-F
Processor	Intel Xeon Gold 6134
Memory	128GB (2x 64GB DDR4 2666MHz ECC server memory)
Hard Drive	2TB - 2x WD 1TB 7200RPM 128MB CACHE 3.5IN SATA Enterprise Class HDD
Optical Drive	24X DVD RW Optical Drive
Display Adapter	Nvidia T400 4GB Graphics Card
PSU	920W redundant PSU



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Item	Description
Server Barebone	Supermicro SuperServer 7048R-C1R Barebone
Processor	2 x Intel Xeon E5-2643 V4 3.40 GHz 6-Core Server Processor
Memory	8 x 64GB DDR4 2400 MHz Load-Reduced Server DIMM
Hard Drive	3 x HGST 1W10001 1TB 7200RPM 128MB CACHE 3.5" SATA Enterprise Class HDD
Optical Drive	24X DVD RW Optical Drive
Display Adapter	NVIDIA Quadro P620 2GB Graphics Card
Network Adapter	Intel X520-DA2 Dual-Port 10GbE Network Adapter
Serial Port	Supermicro 9-PIN Serial Port Cable
	Single Port 16in DB9 Serial Port Bracket
PCI-E USB 3.0	Startech PEXUSB3S44V 4-Port PCIE SuperSpeed USB 3.0 Adapter
Mounting Rails	Supermicro CSE-PT26L 4U Mounting Rails





Main Controller Overview

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To View motherboard information

Click Start > System infomation

Windows 10 LTSC 1809 - Build 17763
Windows 10 LTSC 21H2 - Build 19044

Item	Value
OS Name	Microsoft Windows 8.1 Pro
Version	6.3.9600 Build 9600
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	MY1013
System Manufacturer	Supermicro
System Model	X9SRA/X9SRA-3
System Type	x64-based PC
System SKU	To be filled by O.E.M.
Processor	Intel(R) Xeon(R) CPU E5-2687W 0 @ 3.10GHz, 3101 Mhz, 8 Core(s), 16 Logical ...
BIOS Version/Date	American Megatrends Inc. 3.2, 16-Jan-15
SMBIOS Version	2.7
Embedded Controller Version	255.255
BIOS Mode	Legacy
BaseBoard Manufacturer	Supermicro

Item	Value
OS Name	Microsoft Windows 10 Pro
Version	10.0.18363 Build 18363
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	V810- S2EX
System Manufacturer	Supermicro
System Model	Super Server
System Type	x64-based PC
System SKU	To be filled by O.E.M.
Processor	Intel(R) Xeon(R) Gold 6134 CPU @ 3.20GHz, 3201 Mhz, 8 Core(s), 16 Logical P...
BIOS Version/Date	American Megatrends Inc. 3.4, 11/3/2020
SMBIOS Version	3.2
Embedded Controller Version	255.255
BIOS Mode	Legacy
BaseBoard Manufacturer	Supermicro

ABI Main Controller (Power Supply)

Automated Board Inspection

2140-0023

2140-0069

2140-0092



Power supply LED should turn green once the server boot up.

If power supply LED remain orange, power supply may not plugged properly / defective.

Windows	Server type	HDD Part ID
Window 8.1	Series 2 Rev 3 (motherboard X9SRA)	4XASPP-A124
Window 10 Pro	Series 2 Rev 3 (motherboard X9SRA)	4XASPP-A126
Window 10 Pro	Series 2 Rev 4, 5 (motherboard X11SPL-F)	4XASPP-A127
Window 10 LTSC	Series 2 Rev 3 (motherboard X9SRA)	4XASPP-A365
Window 10 LTSC	Series 2 Rev 4, 5 (motherboard X11SPL-F)	4XASPP-A367
Window 10 LTSC		4XASPP-A366

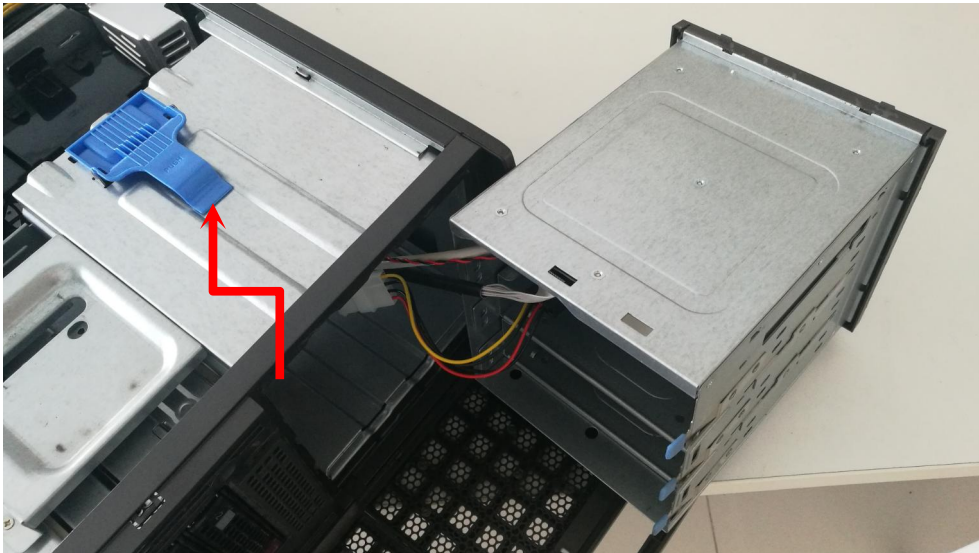


Main Controller Overview

Automated Board Inspection

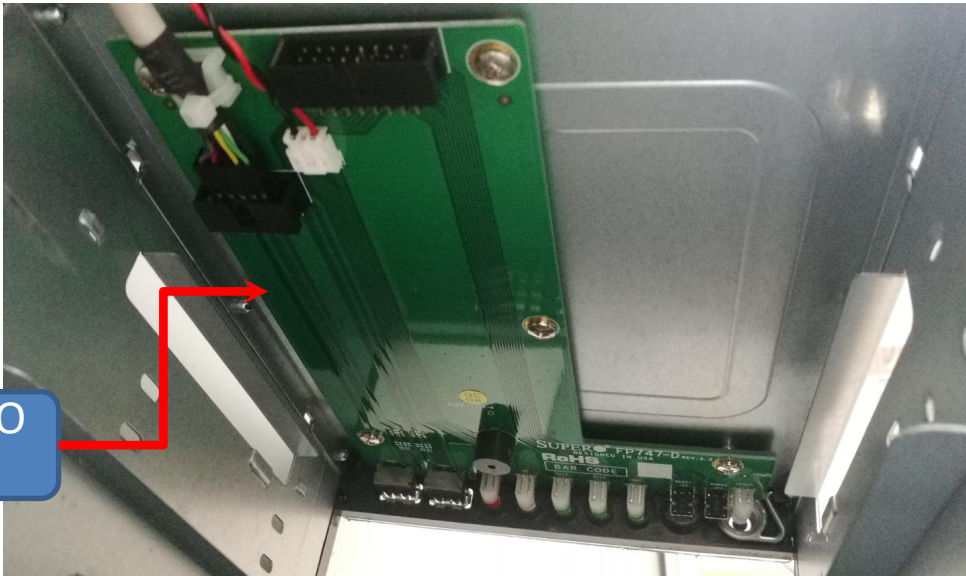
Pull to Unlock

Chassis



DVD ROM

Front Panel IO card



ABI Graphic Card - GPU

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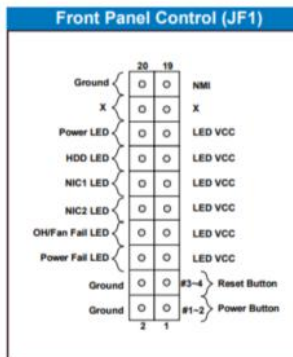
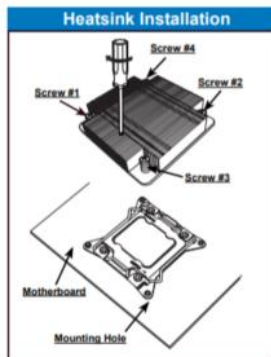
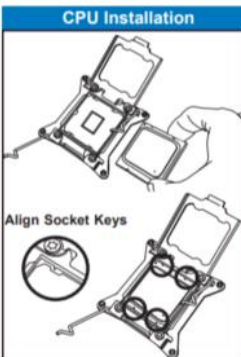
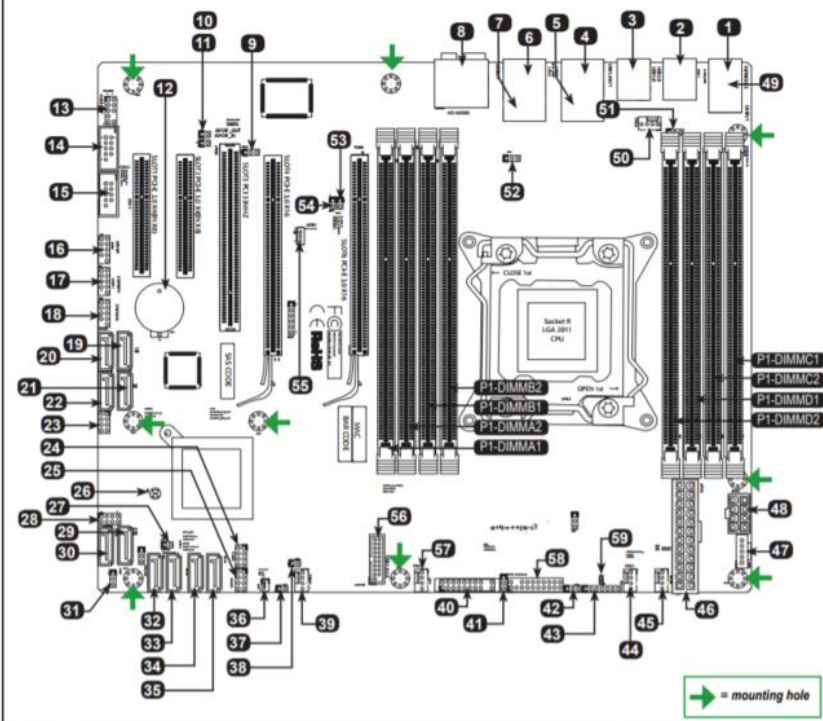


Basic Package



3DCT Lite Package

Motherboard Layout and Features



Jumpers, Connectors and LED Indicators

Jumpers			
9	JPAC1	Audio Enable	Pins 1-2 (Enabled)
31	JPME1	Intel ME Mode Select	Pins 1-2 (Normal)
26	JBT1	CMOS Clear	Short contact pads to reset CMOS
37	JCF1	CF Card Master/Slave Select	On (Master)
41	JPT1	TPM Enable	Pins 1-2 (Enabled)
42	JWD1	Watch Dog Reset	Pins 1-2 (Reset)
51	JPUS1	USB Wake-up Enable	Pins 1-2 (Enabled)
52	JPL2	LAN2 Enable/Disable	Pins 1-2 (Enabled)
53,54	J12C2/J12C1	SMB to PCI Slots	On (Enabled)

Connectors		
1	JKBMS1	Combination PS/2 Keyboard/Mouse Port
2,3,7,49	USB 0/1, 2/3, 4/5, 6/7, 0/1	Back Panel USB 2.0 Ports
4,6	LAN1, LAN2	Gb Ethernet LAN Ports (LAN1 & LAN2)
5	USB 3.0 0/1	Back Panel USB 3.0 Ports
8	HD Audio	High Definition (HD) Audio output jacks
10,11	JSPDIF OUT/IN	SP/DIF Audio In/Out Headers
12	BT1	System Battery
13	AUDIO FP	Front Panel Audio Header
14,15	COM2, COM1	COM1 & COM2 Serial Port Headers
16,17,18	USB 8/9, 10/11, 12/13	USB 2.0 Headers for front panel access
19,20,21,22	SAS0-SAS3	SATA 2.0 (3Gb/sec)
23,28	3-SGPI01, 3-SGPI02	Serial General Purpose I/O Headers for SAS
24,25	T-SGPI02, T-SGPI01	Serial Link General Purpose I/O Headers
27	JL1	Chassis Intrusion Header
29,30	I-SATA1, I-SATA0	Serial ATA ports (SATA 3.0), 6Gb/s
32,33,34,35	I-SATA2-5	Serial ATA ports (SATA 2.0), 3Gb/s
36	JWF1	SATA DOM (Disk On Module) Power Connector
38	JOH1	Overheat LED/Fan Fail
39,45,44,50,57	FAN A,1,2,4,3	Internal Fan Headers
40	JF1	Front Panel Control Header
43	JD1	Power LED / Speaker Header (Pins 4-7: External Speaker)
46	JPW1	24-pin Main ATX Power Connector
47	JPI2C1	Power Supply SMBus I2C Header
48	JPW2	8-pin Secondary Power Connector
55	JSTBY	Legacy Wake On LAN Header
56	JUSB2/3 (USB 3.0)	USB 3.0 Header for USB 2/3
58	JTPM1	Trusted Platform Module (TPM) Header

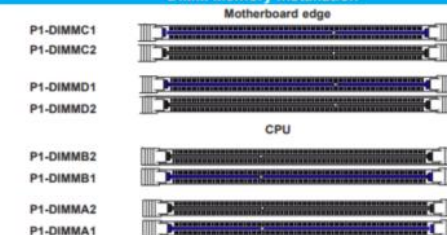
LED Indicators		
59	DP2	Power On LED
		Green: Solid On System is On

Memory Support

The X9SRA motherboard series supports up to 256GB of 1600/1066/1333 MHz ECC/Non-ECC DDR3 DIMMs in eight (8) memory slots (UDIMM/RDIMM).

Note: For memory optimization, use only DIMM modules that have been validated by Supramicro. For the latest memory updates, please refer to our website at <http://www.supramicro.com/products/motherboard>.

DIMM Memory Installation



Memory Population Guidelines

When installing memory modules, the DIMM slots should be populated in the following order: P1-DIMMA1, P1-DIMMB1, P1-DIMMC1, P1-DIMMD1 then P1-DIMMA2, P1-DIMMB2, P1-DIMMC2, P1-DIMMD2.

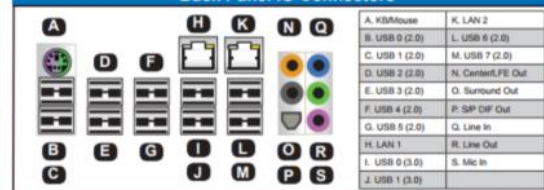
- Always use DDR3 DIMM modules of the same size, type and speed.
- Mixed DIMM speeds can be installed. However, all DIMMs will run at the speed of the slowest DIMM.
- The motherboard will support odd-numbered modules installed (1, 3, 5, or 7 modules). However, for best memory performance, install DIMM modules in pairs.

Recommended Population (Balanced)

P1-DIMMA1	P1-DIMMB1	P1-DIMMC1	P1-DIMMD1	P1-DIMMA2	P1-DIMMB2	P1-DIMMC2	P1-DIMMD2	Total
2GB	2GB							4GB
2GB	2GB	2GB	2GB					8GB
2GB	2GB	2GB	2GB	2GB	2GB			12GB
2GB	2GB	2GB	2GB	2GB	2GB	2GB		16GB
4GB	4GB							8GB
4GB	4GB	4GB	4GB					16GB
4GB	4GB	4GB	4GB	4GB	4GB			24GB
4GB	4GB	4GB	4GB	4GB	4GB	4GB	4GB	32GB

Note: Up to 256GB of memory are supported. See chapter 2 of the User Manual for complete memory population information.

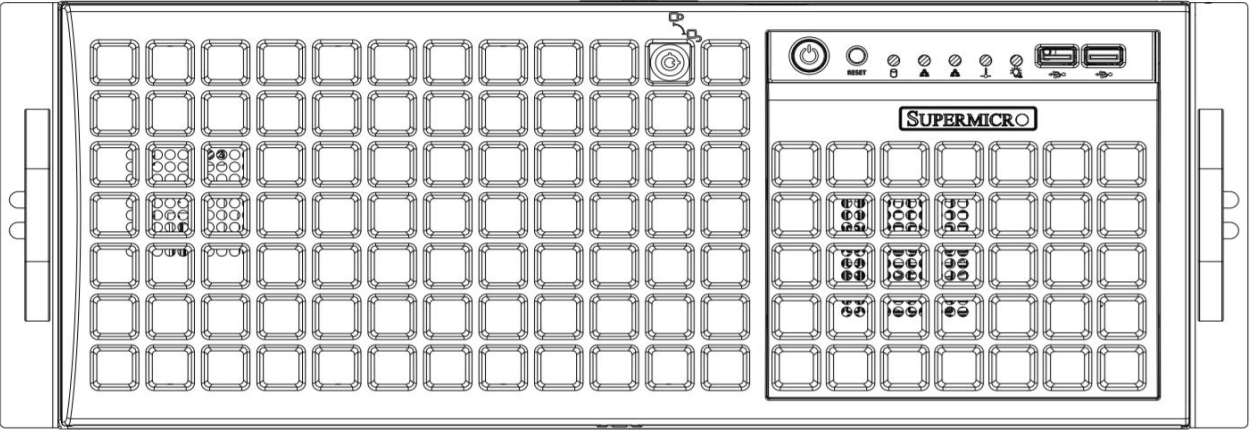
Back Panel IO Connectors





Control Panel Buttons and Indicator

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Indicator	Type	Description
	Power	The main power switch is used to apply or remove power from the power supply to the server system. Turning off the power with this button removes the main power but keeps standby power supplied to the system. Therefore, you must unplug system before servicing.
	Reset	The reset button is used to reboot the system.
	HDD	Indicates SAS/SATA drive, and/or DVD-ROM drive activity when flashing

Indicator	Type	Description
	NIC1	Indicates network activity on GLAN1 when flashing.
	NIC2	Indicates network activity on GLAN2 when flashing.
	Information	Solid Red : Overheat Blink Red (quick) : Fan fail Blink Red (slow) : Power fail Solid Blue : Local UID Blink Blue (quick): Remote UID
	Power Fail	Indicates a power failure to the system's power supply units.

Part ID		2160-0049	2160-0149	2160-0187
Photo (Front)				
Photo (Rear)				
Power adapter Part ID		2140-0082	2140-0094	
Remark	obsolete	obsolete	Current (For XMP, ZMP Mei Card)	Current (Vicat IO, Panasonic drive)



Remote Motion Server (New RMS/Atom PC)

Automated Board Inspection

For atom pc upgrade to 2160-0149, new slider kit required

	Applicable machine
4XASPP-A256	STD, S2EX, XXL
4XASPP-A257	XLT

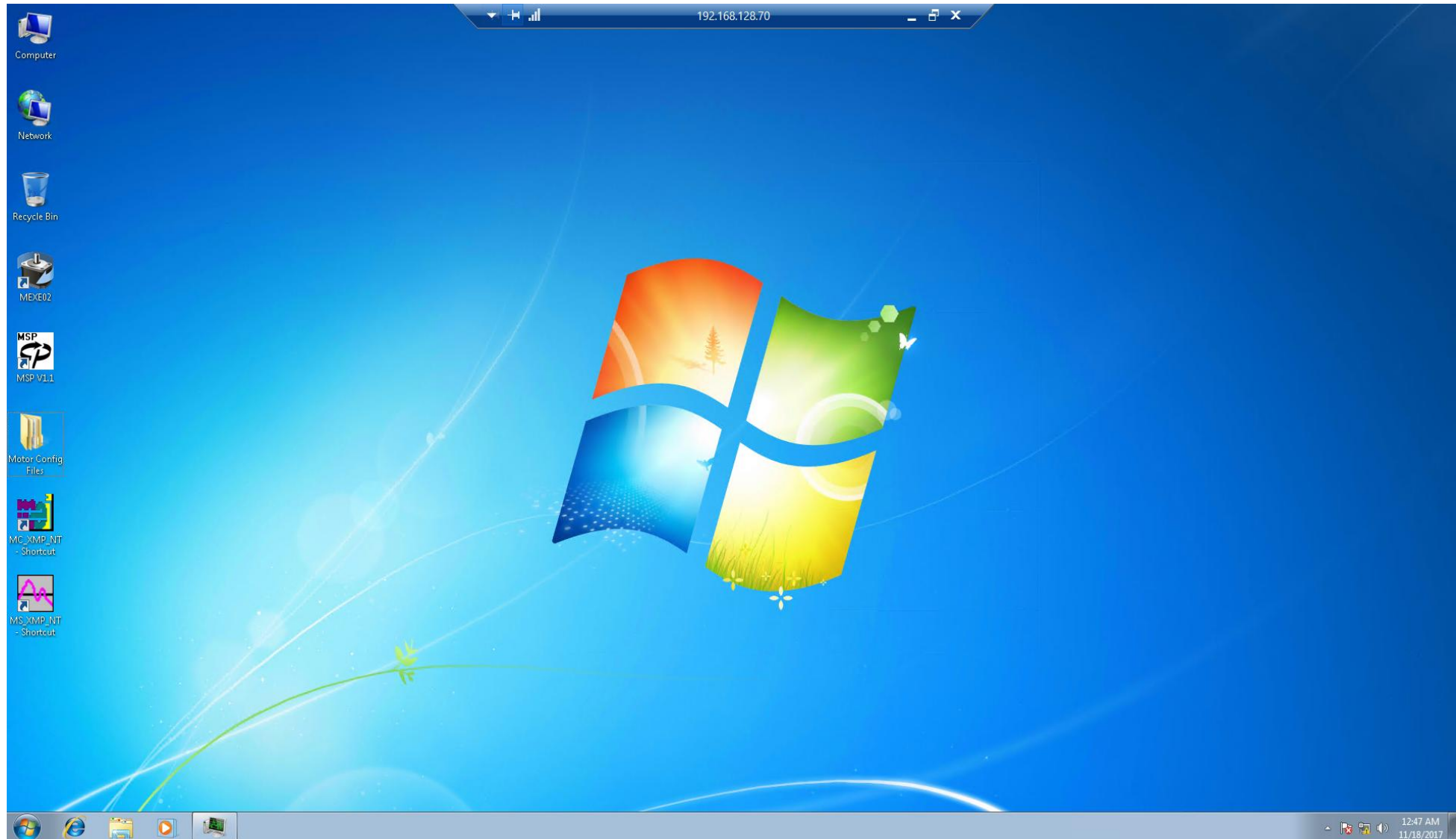
For atom pc upgrade / replacement, MEI card required to transfer over





Automated Board Inspection

Remote Motion Server (RMS/Atom PC) - Interface





Gigabit Switch - Specification

Automated Board Inspection



	Black Cisco Switch
Part ID	2150-0172
Username	Cisco
Password	Cisco
Colour	Black



	White Cisco Switch
Part ID	2190-0006
Username	Admin
Password	Cisco
Colour	White
Teaming	No supported



Cisco Switch - Upgrade

Automated Board Inspection

For Cisco switch upgrade to white Cisco switch, new slider kit required

Part ID	applicable machine
4XASPP-A264	S2EX,STD,XLT
4XASPP-A265	XXL

For machine equipped with Windows 8 server, network card required to be replaced if upgrade white Cisco Switch

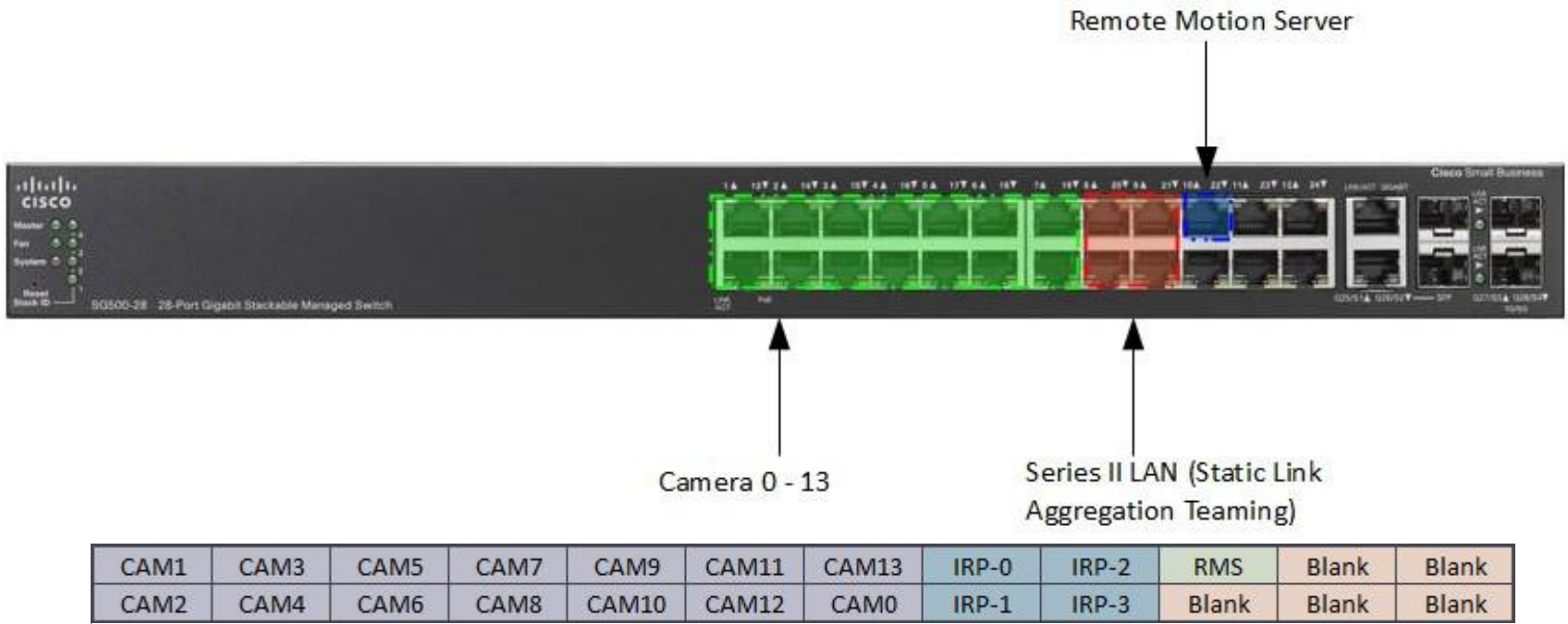
Part ID	Description
2511-0017	DAC Cable
2150-0131	X520 Network card





Cisco Switch Port Layout (Black)

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LAN Port Connectivity Layout (MUST follow Orientation)

User Name : cisco
Password : cisco
IP Address : 192.168.128.122
Subnet Mask : 255.255.255.0

Port GE8, GE9, GE20 and GE21 assign as teaming port to increase bandwidth for data transferring.
(for windows 8 machine)



Cisco Switch Port Layout (White)

Automated Board Inspection



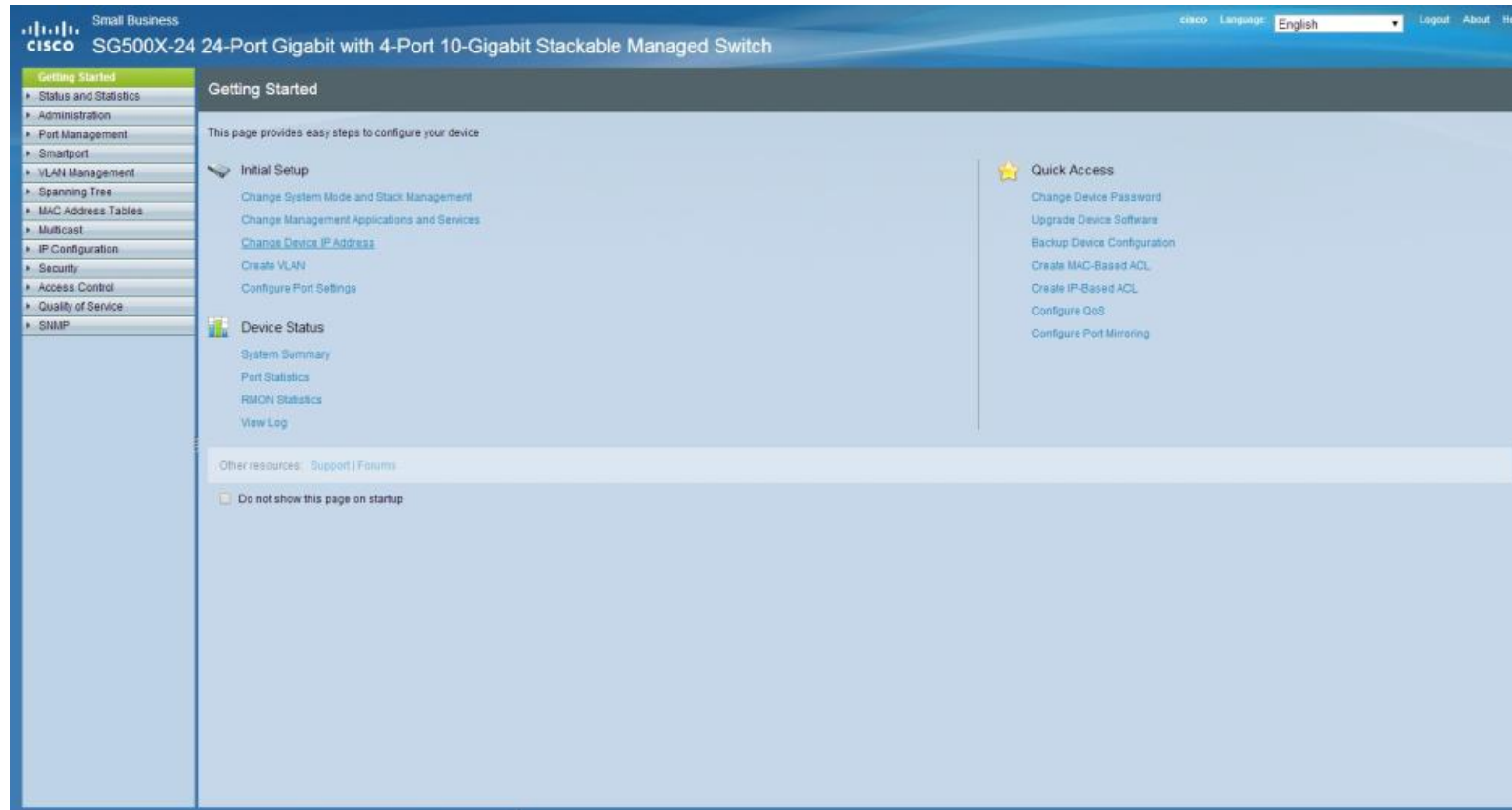
CAM 1	CAM 3	CAM 5	CAM 7	CAM 9	CAM 11
CAM 2	CAM 4	CAM 6	CAM 8	CAM 10	CAM 12

CAM 13	Vicat IO	Atom PC	Blank	Blank	Blank
CAM 0	Moons	Blank	Blank	Blank	Blank

server			
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LAN Port Connectivity Layout (MUST follow Orientation)

User Name : admin
Password : cisco
IP Address : 192.168.128.122
Subnet Mask : 255.255.255.0



- Enable Jumbo Frames. (Under Port Management → Port Settings)
- Disable STP for LAG1. (Under Spanning Tree → STP Interface Settings)
- GE8, GE9, GE20 and GE21 assign as ToSingleServer team member. (Under Port Management → Link Aggregation → LAG Management → LAG1)
- Load Balance Algorithm set as IP/MAC Address (Port Management → Link Aggregation → LAG Management)



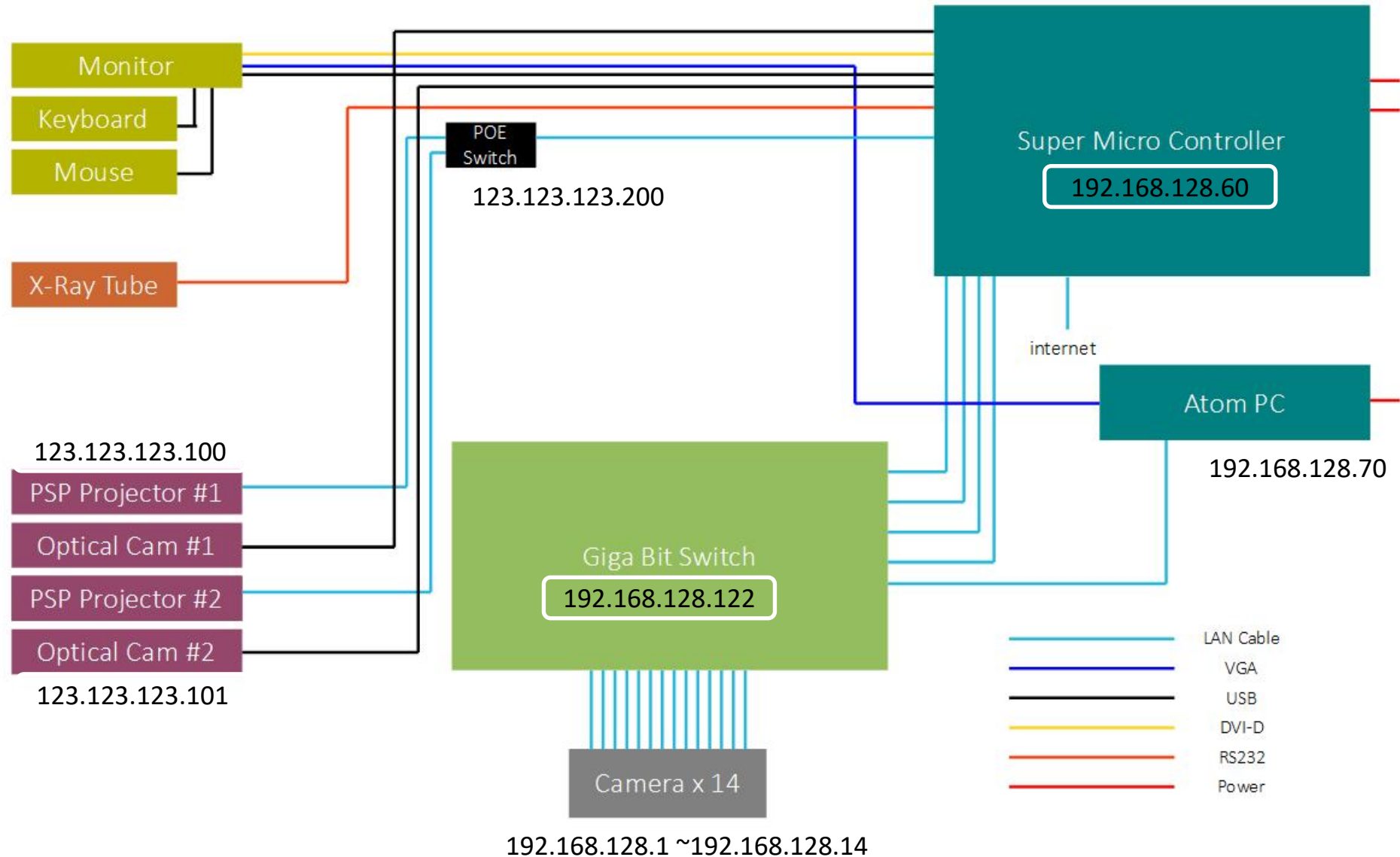
Network Card

Automated Board Inspection

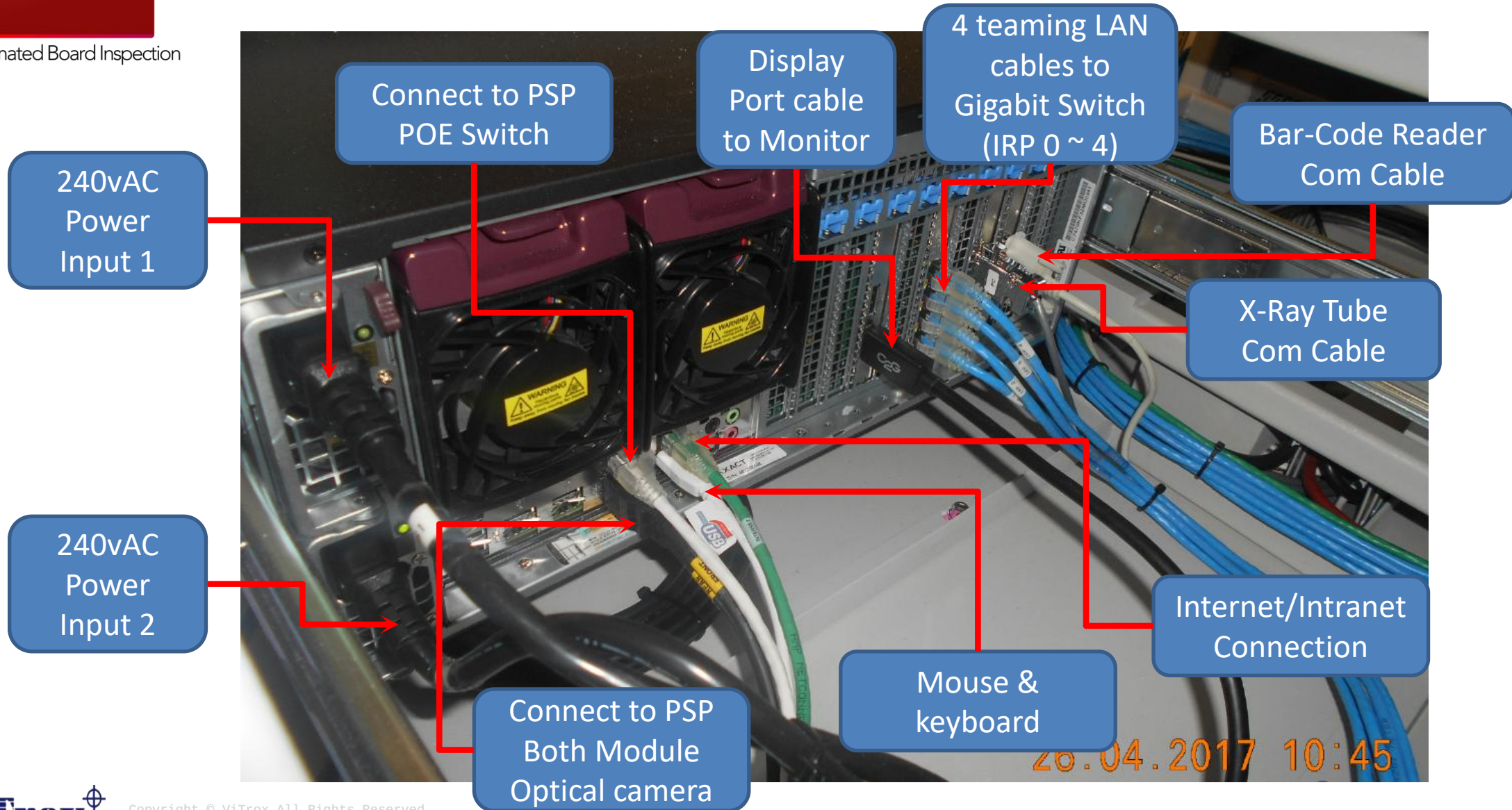
Part ID	2172-0006	2150-0131
Discription	HOST ADAPTER,RJ45 QUAD PORTS, PCI-E X4	HOST ADAPTER,SFP+ DIRECT ATTACH COPPER; NETWORK 2 PORTS CARD 1GBE / 10GBE,PCI-E X8 GEN2
Applicable machine type	Used for machine equipped with Windows 8	Used for machine equipped with Windows 10
Remark	Teaming required	

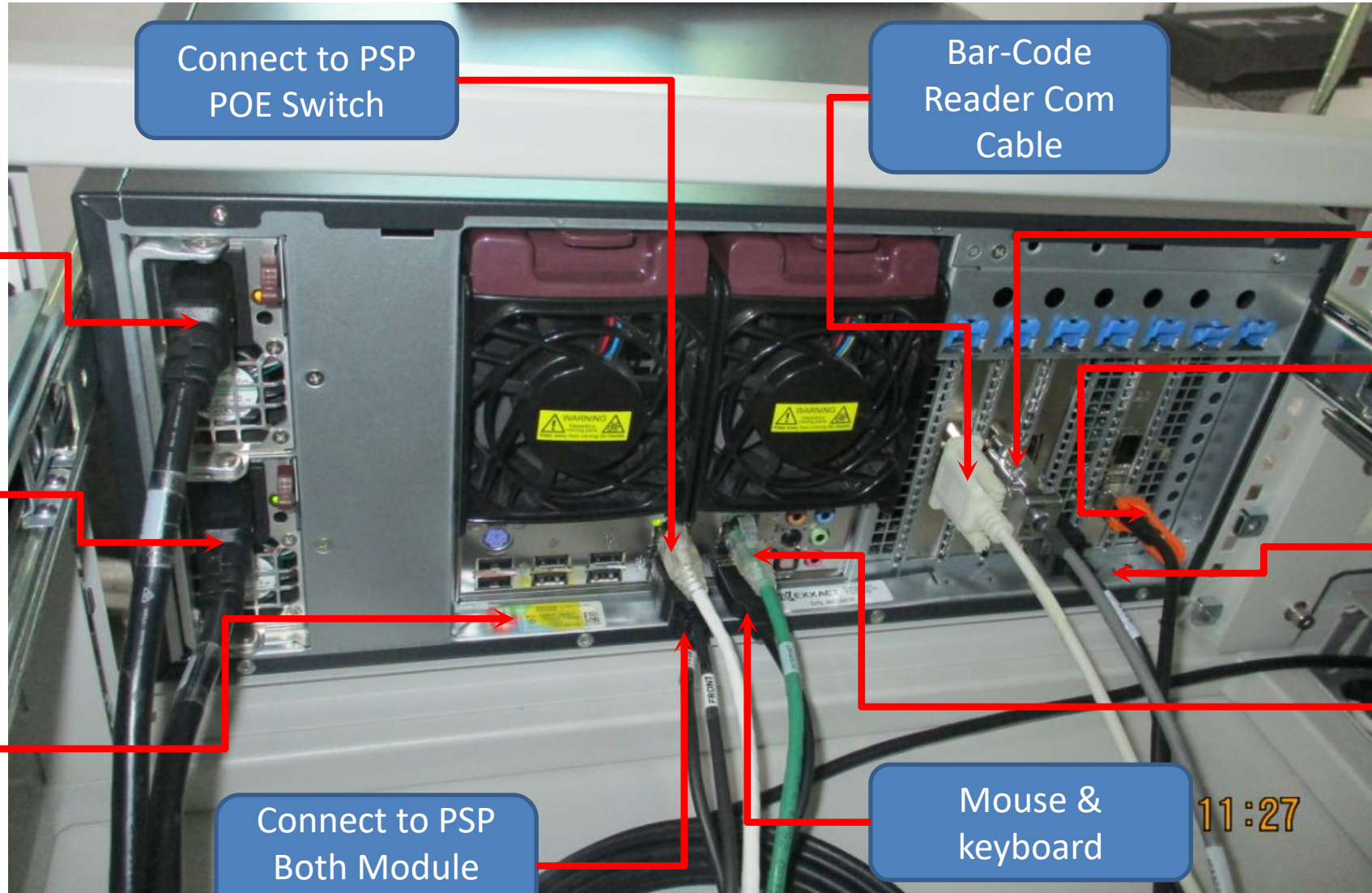


Automated Board Inspection



Hardware Overview - Main Controller (Win 8)





240vAC
Power
Input 1

240vAC
Power
Input 2

Window
OS Key

Connect to PSP
POE Switch

Connect to PSP
Both Module
Optical camera

Bar-Code
Reader Com
Cable

Mouse &
keyboard

X-Ray Tube
Com Cable

SFP Cable
(Win 10 Only)

Display Port
cable to
Monitor

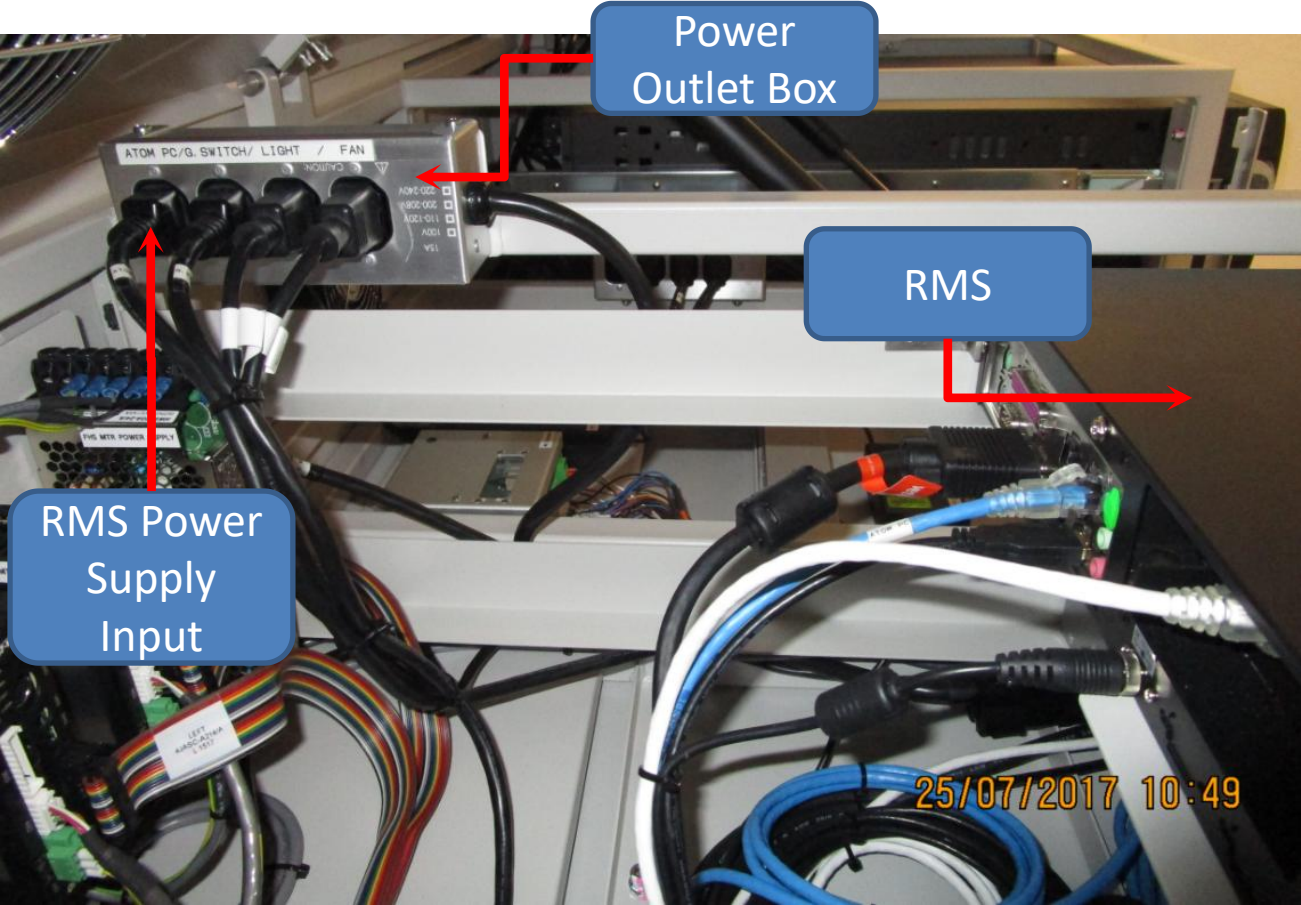
Internet/Intranet
Connection

11:27



Hardware Overview - RMS (Atom PC)

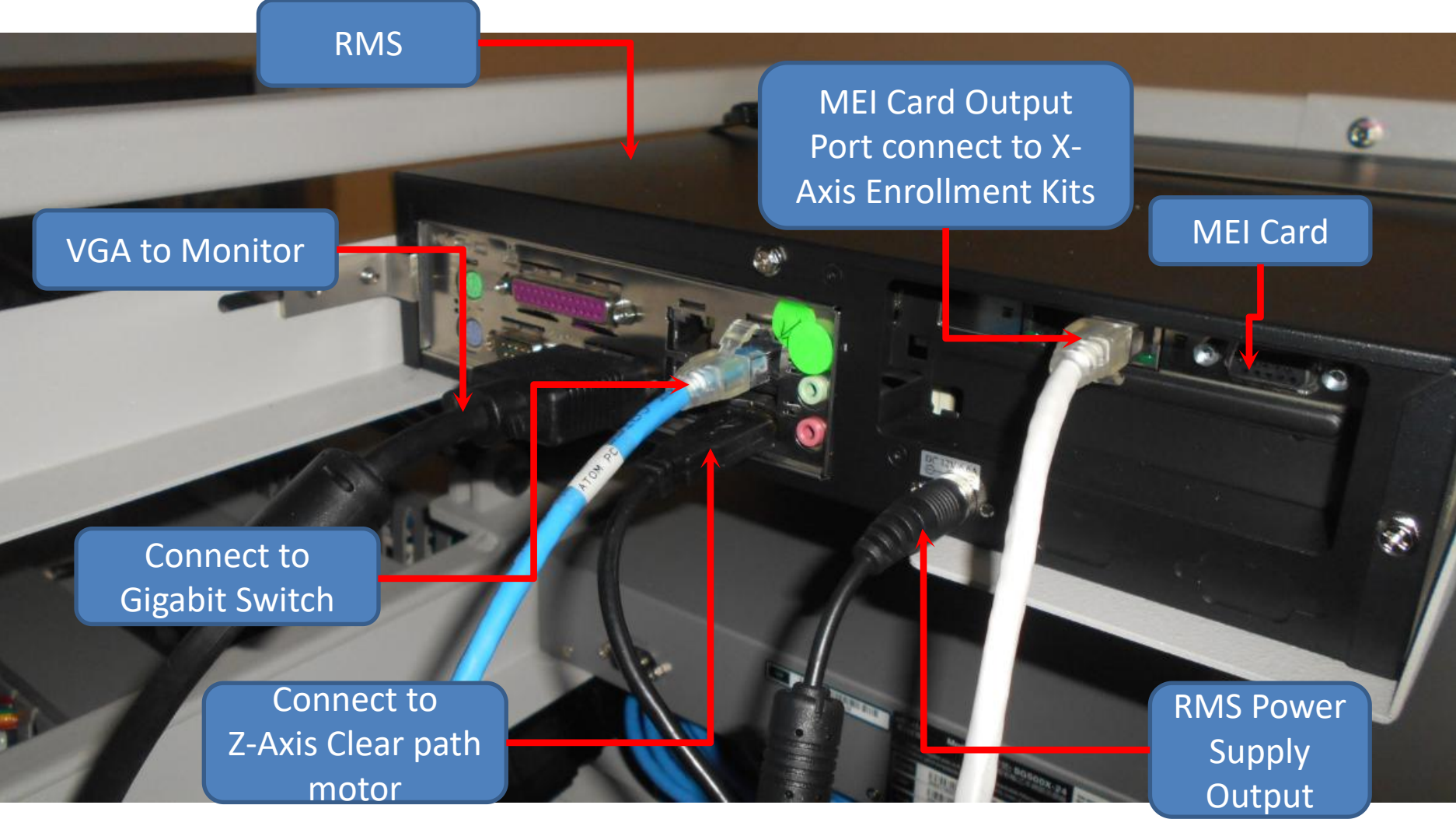
Automated Board Inspection



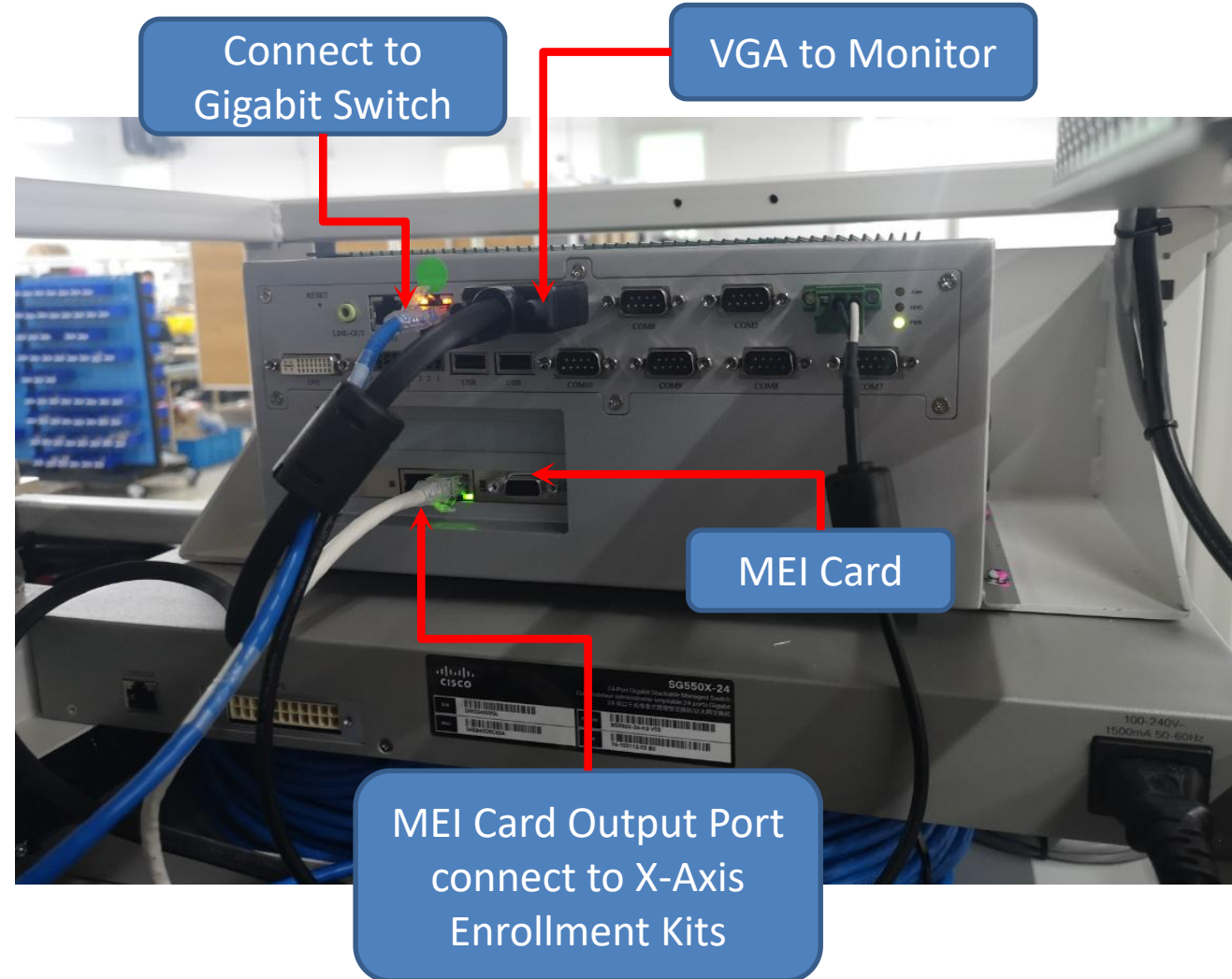
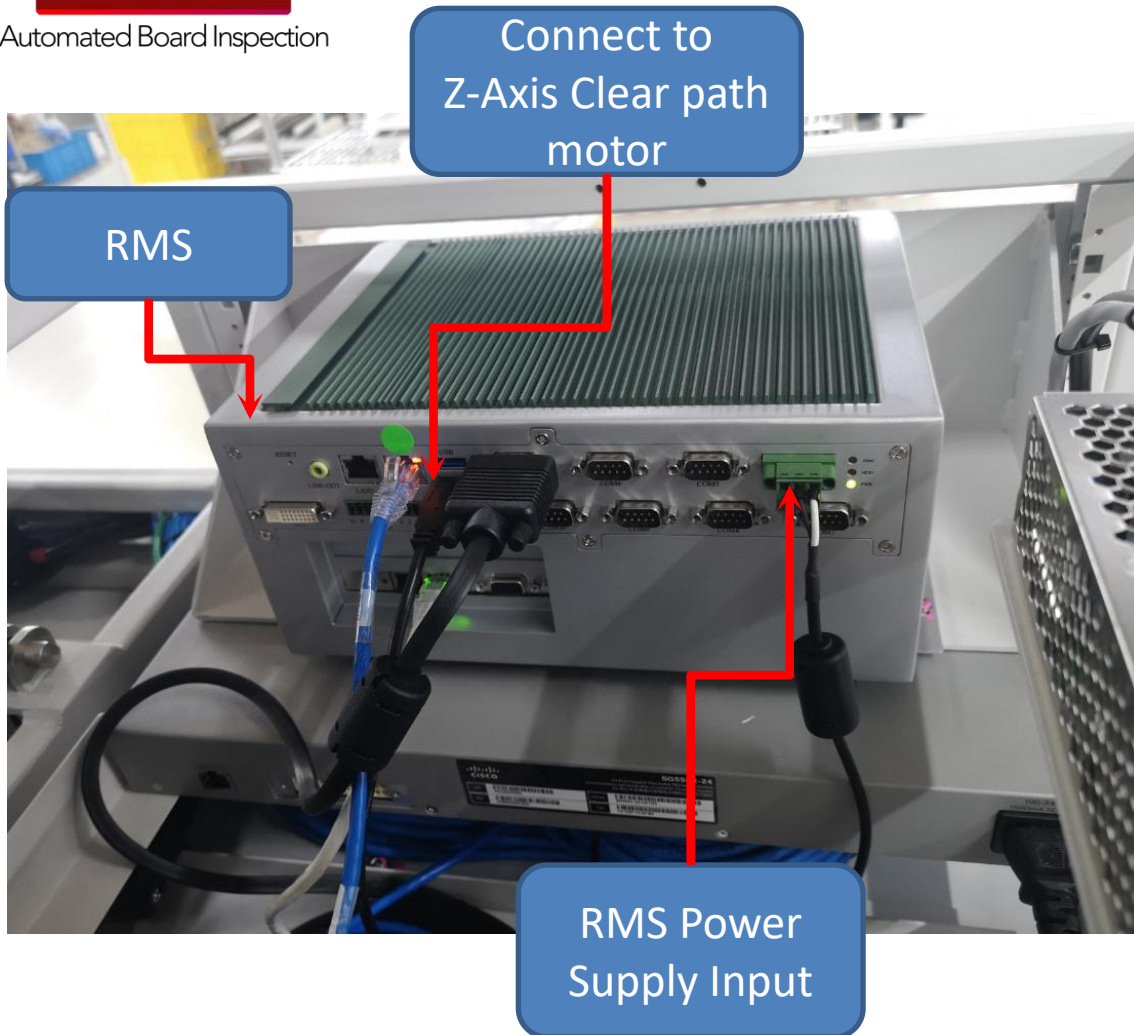


Hardware Overview - RMS (Atom PC)

Automated Board Inspection

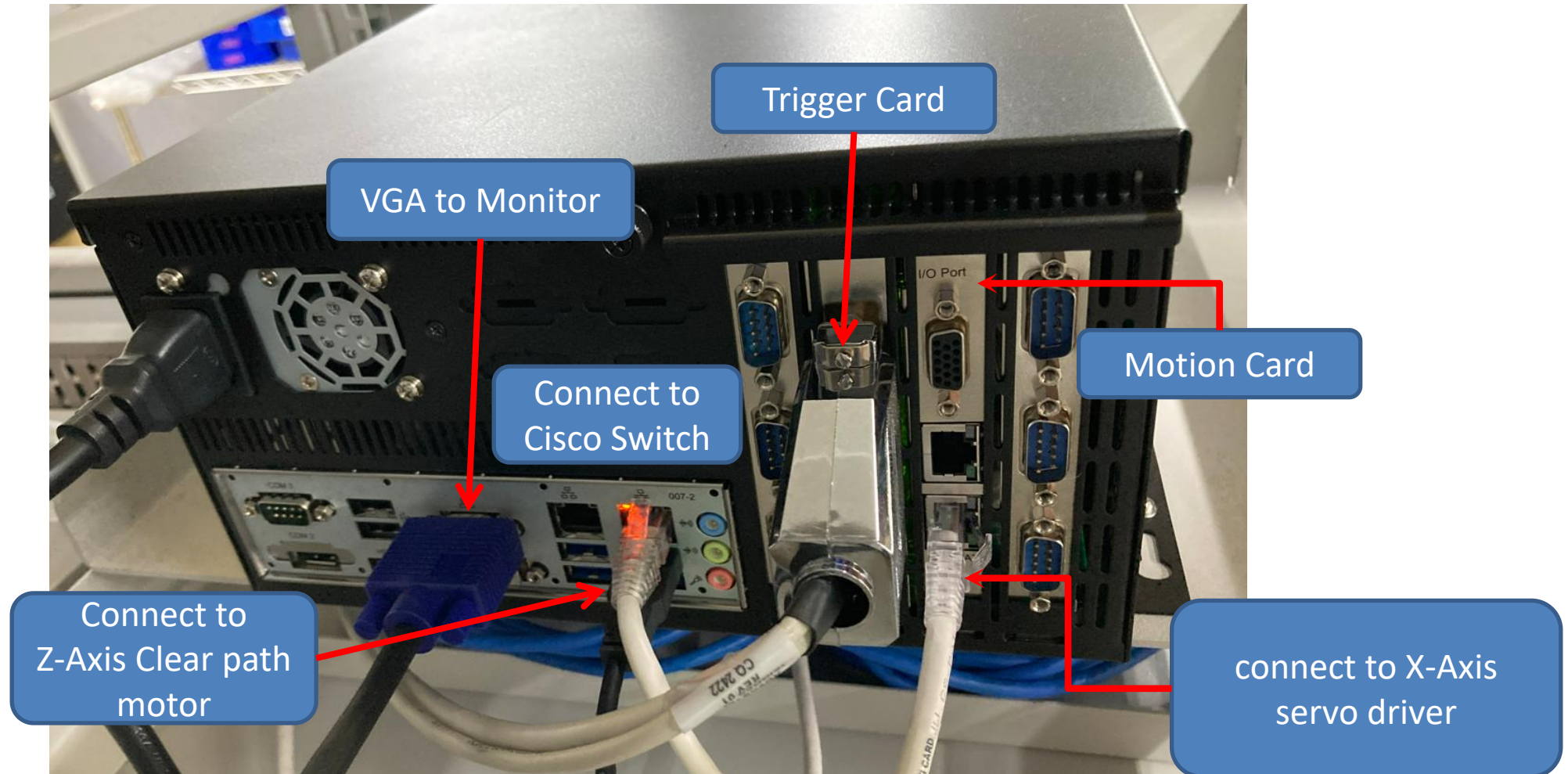


Automated Board Inspection



ABI Hardware Overview - RMS (Atom PC for Panasonic drive)

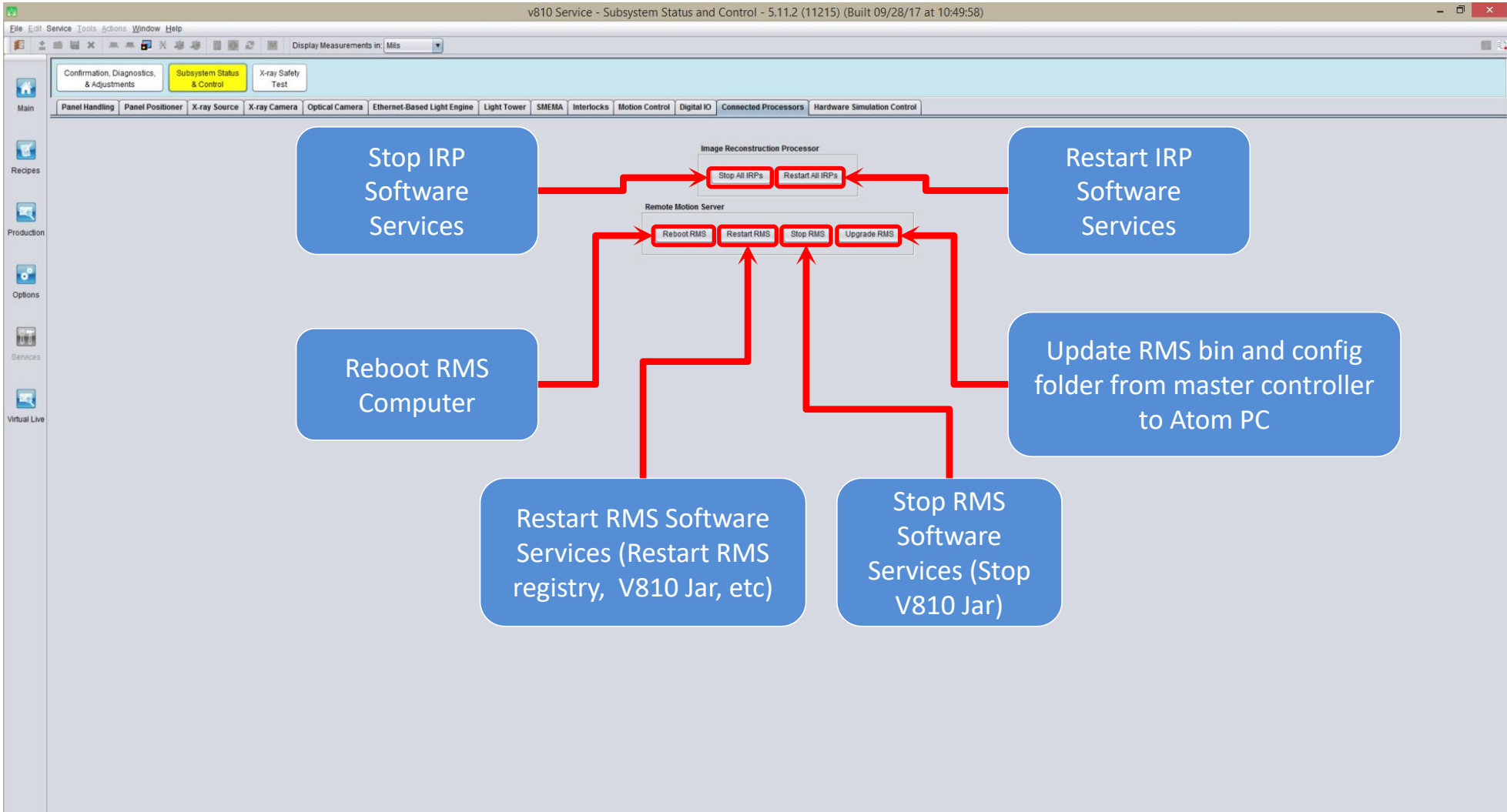
Automated Board Inspection





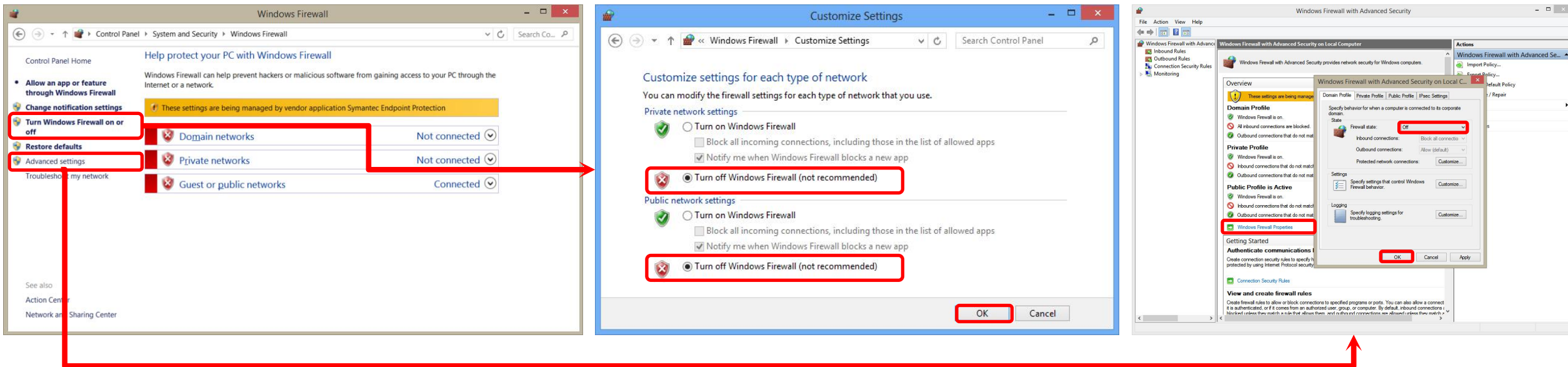
V810i GUI Connected Processor Main Tab

Automated Board Inspection

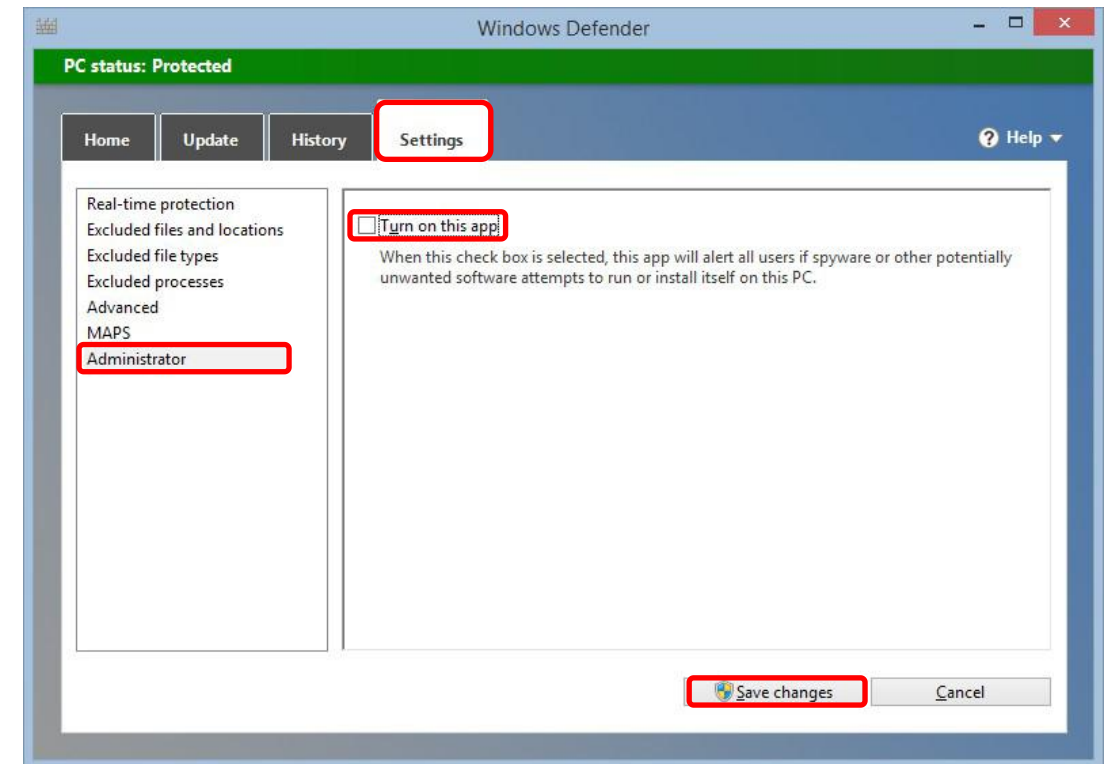
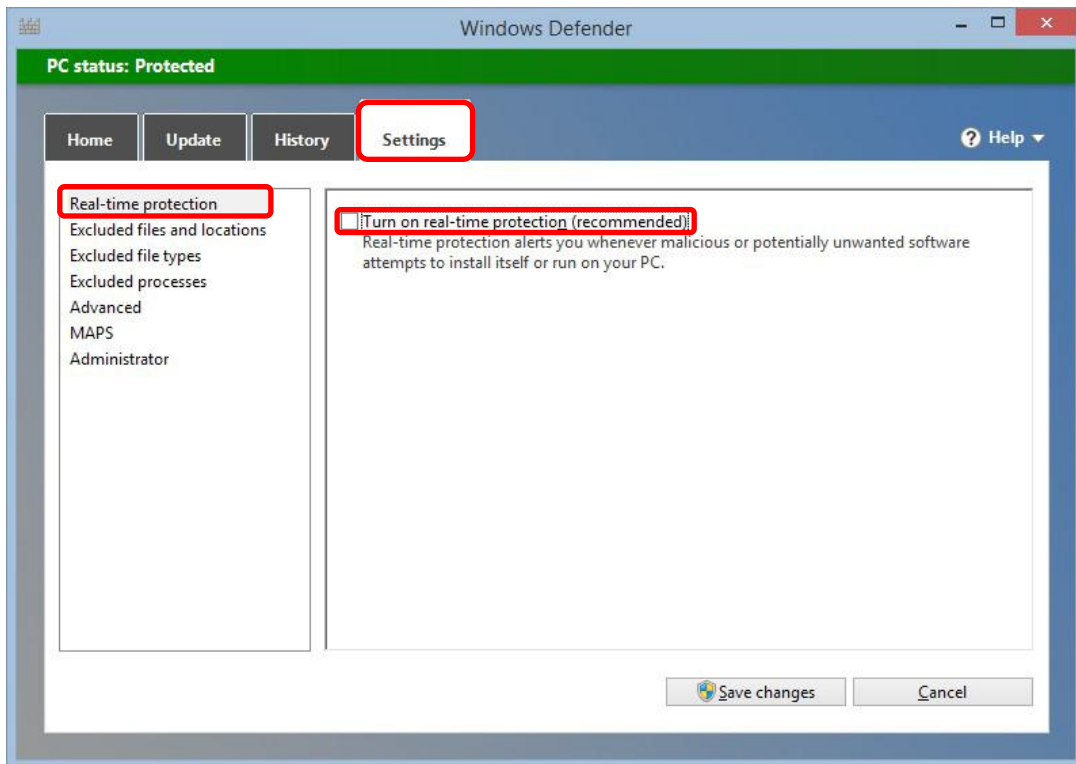


ABI Controller Server Default Setting

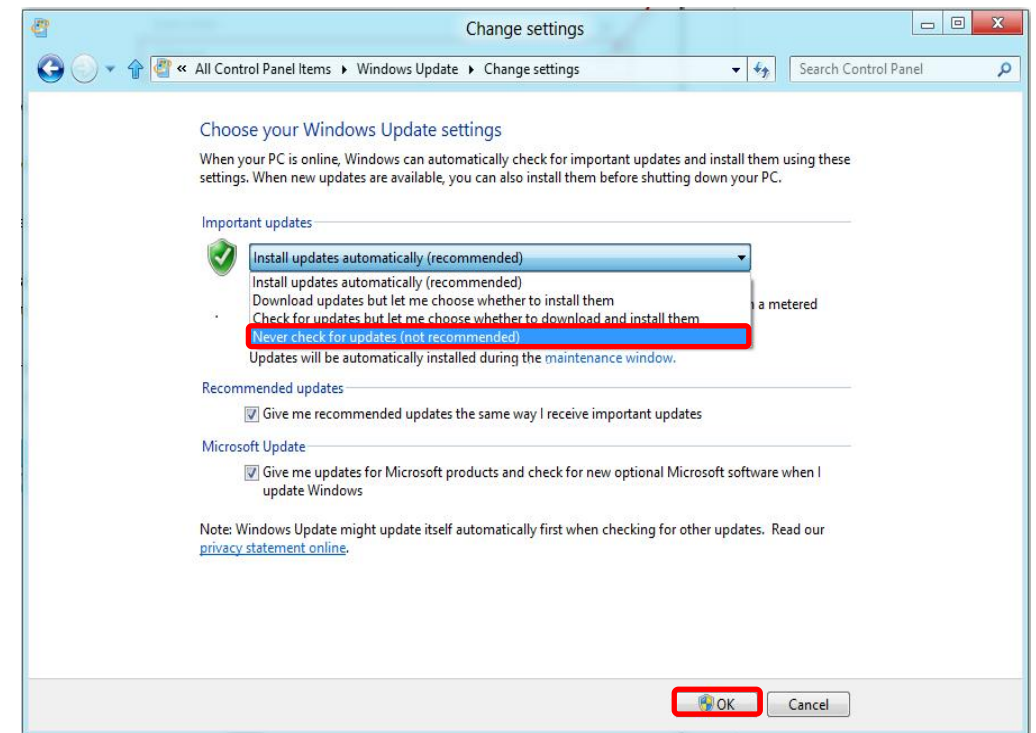
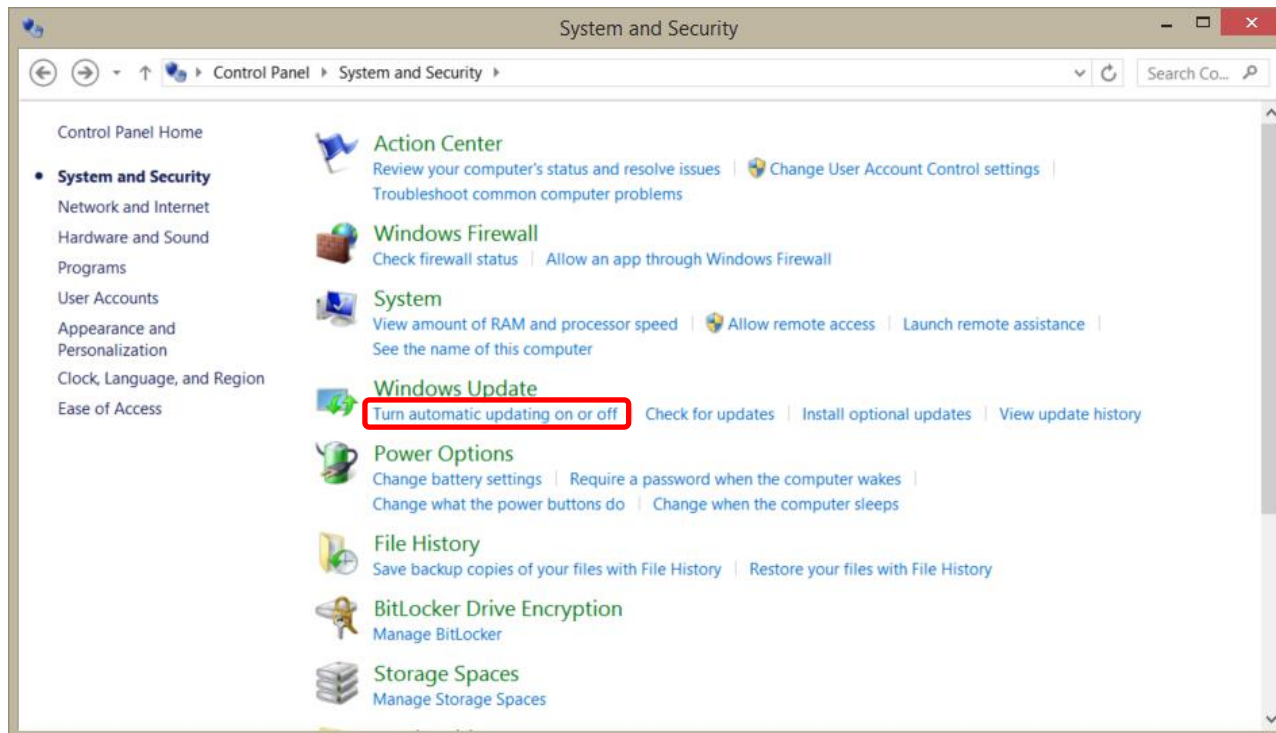
Automated Board Inspection



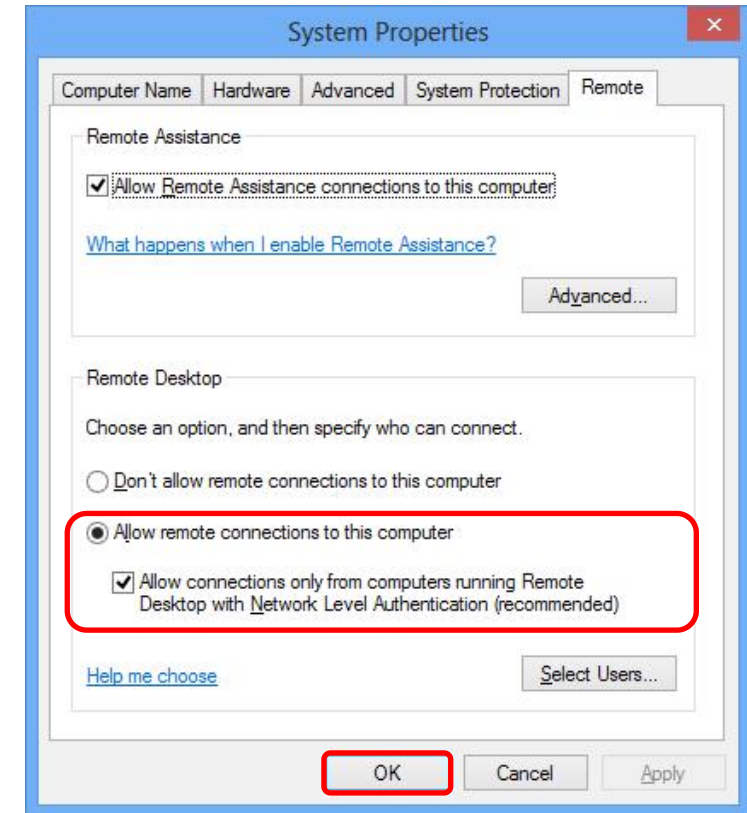
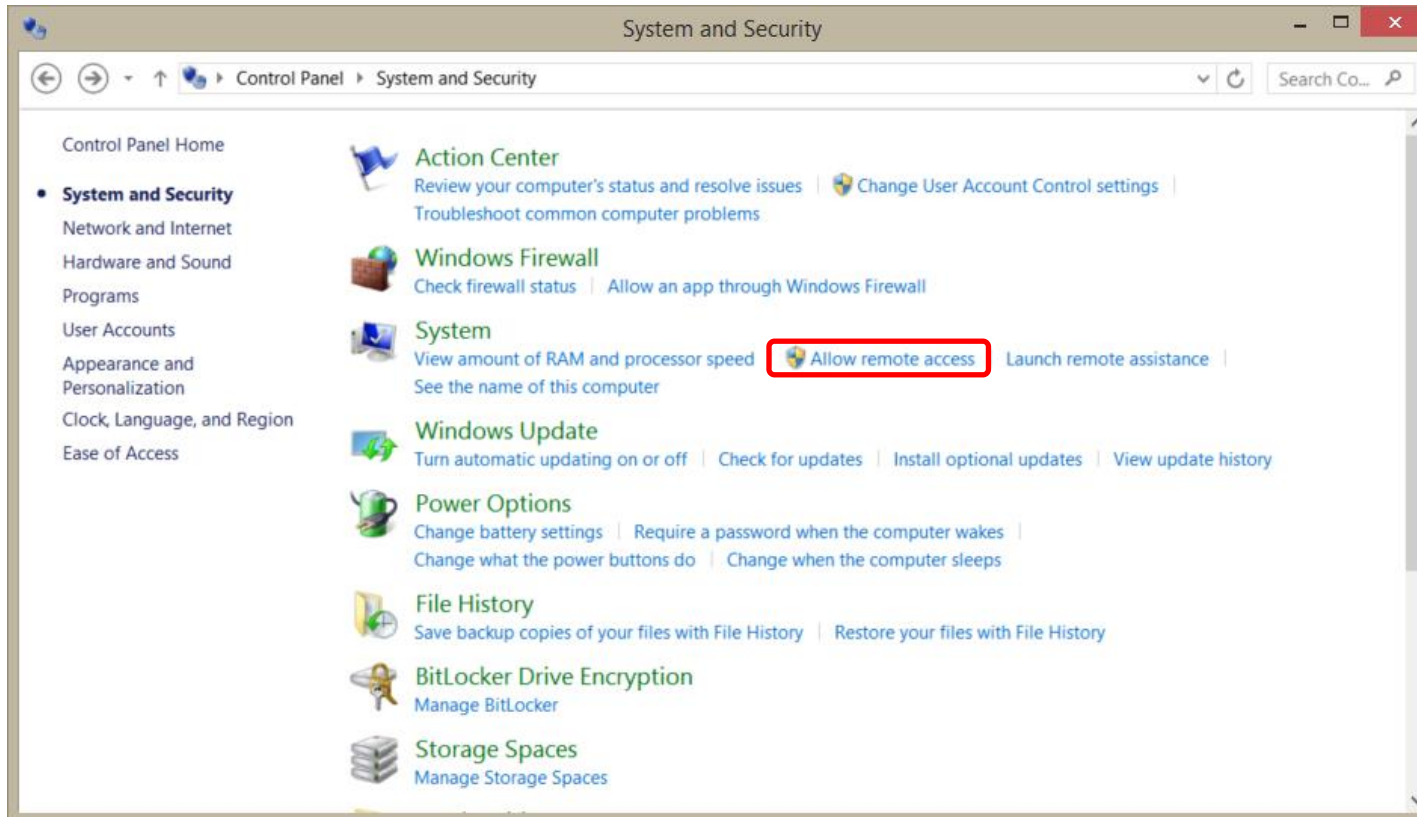
Disable Window Firewall at Control Panel



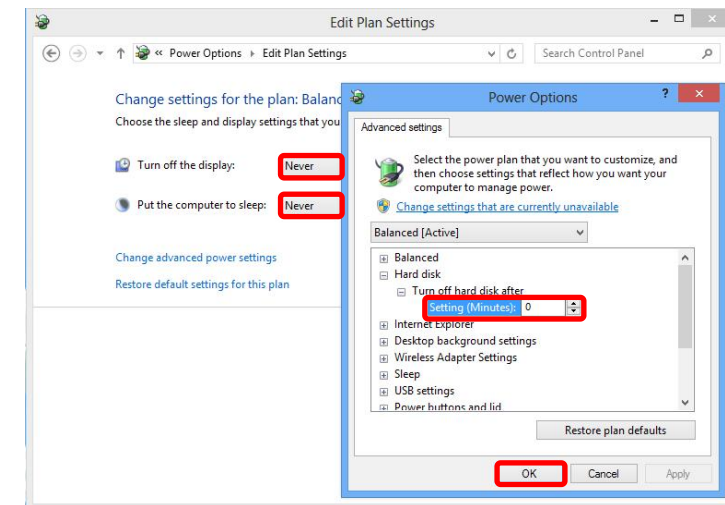
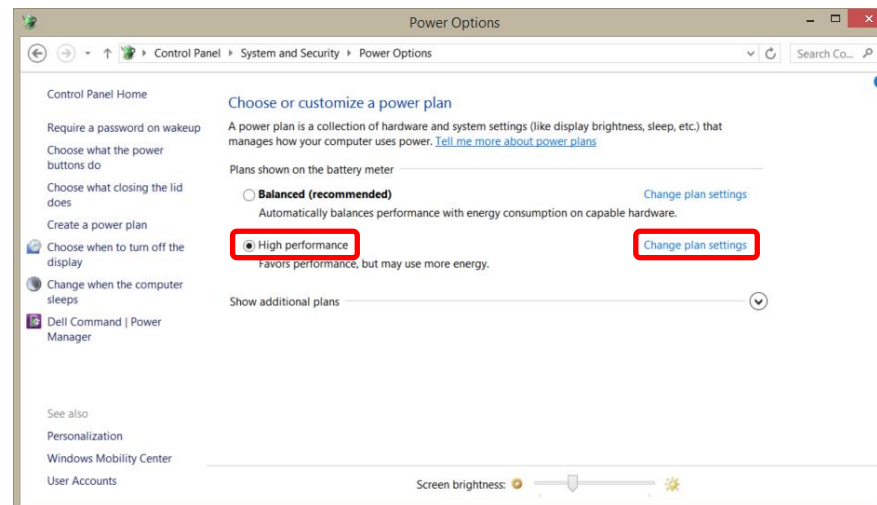
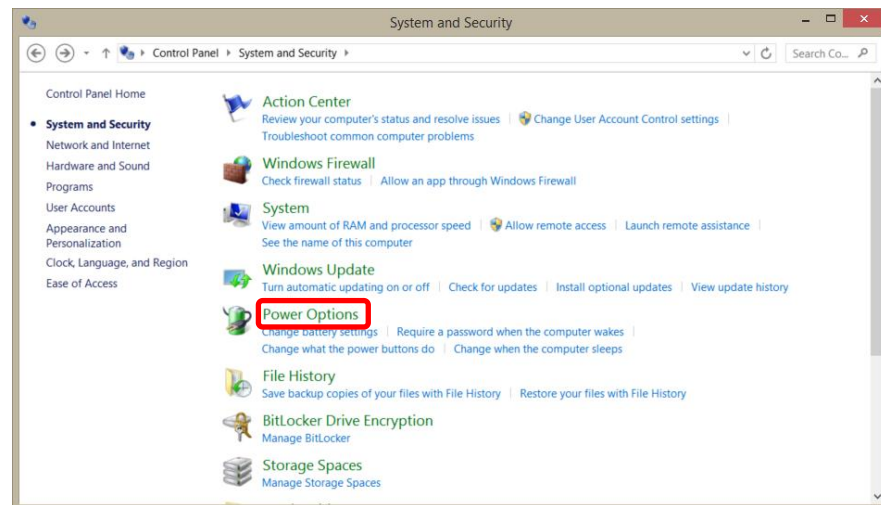
Disable Window Defender at Control Panel



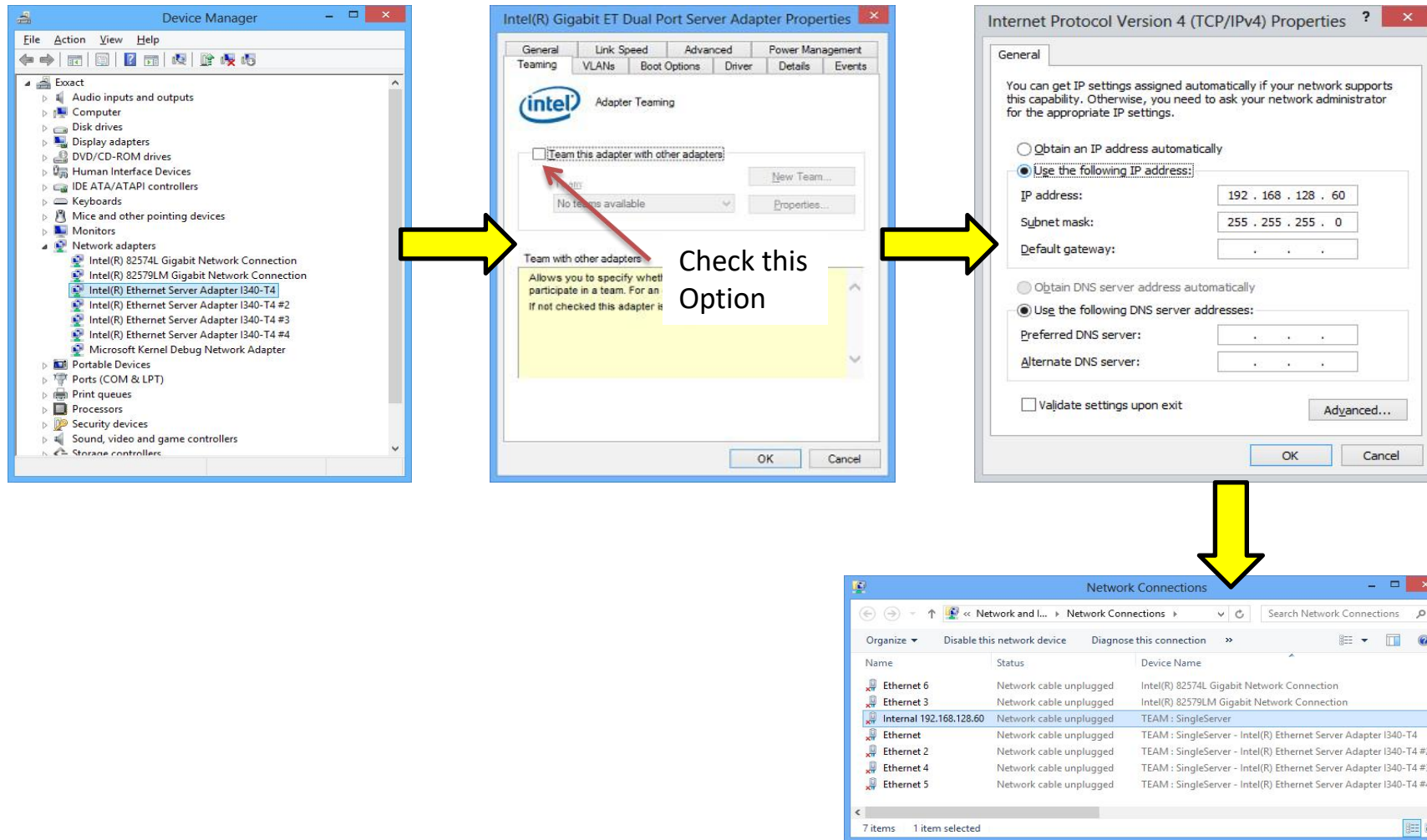
Disable Window Update at Control Panel



Enable allow Remote Desktop setting



Enable High Performance Power Option





Automated Board Inspection

Main Controller Server IP Config Setting

- Connect Super Micro Server with Gigabyte Switch

Super Micro Server → External 4P Network card → Cisco Gigabit Switch

- LAN port run with function Network Teaming

(Bandwidth aggregation, data transfer more stable and faster.

Prevent connectivity loss in the event of a network component failure).

- IP setting below at Internet Protocol Version 4 (TCP/IPv4) at

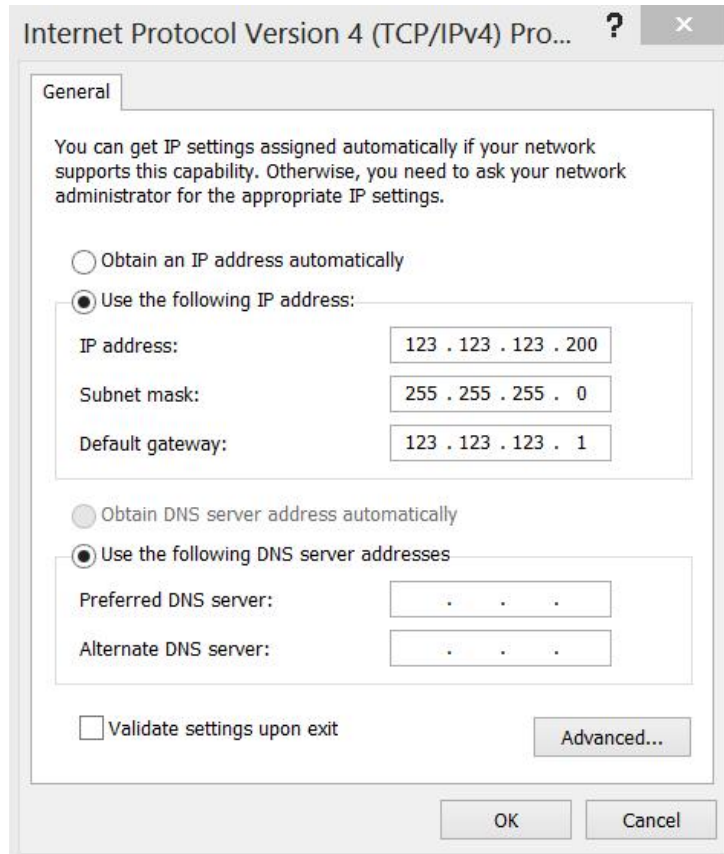
Micro server-Internal LAN port.

➤ IP Address: 192.168.128.60

➤ Subnet Mask: 255.255.255.0

➤ Leave the other settings blank

Once done, need rename LAN Port Icon to Internal 192.168.128.60 (easy to identify).



- PSP2 is new feature which used LAN port connect to MC and PSP Projector

Super Micro Server → POE Switch → PSP Projector
(Front/Rear)

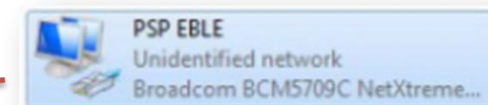
- IP setting below at Internet Protocol Version 4
(TCP/IPv4), (Micro server -Internal LAN port)

- IP Address: **123.123.123.200**

- Subnet Mask: **255.255.255.0**

- Default gateway: **123.123.123.1**

Once done, need rename LAN Port Icon to **PSP EBLE**
(easy to identify)



ABI RMS (ATOM PC) IP Config Setting

Automated Board Inspection

	v810	v810 Series II
IP Address	10.1.1.55	192.168.128.70
Subnet Mask	255.0.0.0	255.255.255.0
Default Gateway	10.1.1.254	-
Preferred DNS Server	10.1.1.254	-

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

Subnet mask:

Default gateway:

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

V810 RMS IP Address Setting



☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

Subnet mask:

Default gateway:

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

V810 Series II RMS IP Address Setting

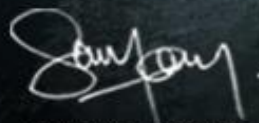
Hardware Overview - Main Controller With 3DCT Advance Kits



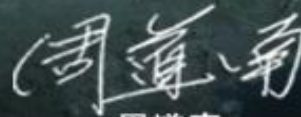
Connect 3DCT
Advance Kit
Server PC



WEE KAH KHIM
Vice President & General Manager



GARY LEONG
Sales Director



周道南
Sales General Manager
(China & Taiwan)



SEOW ZI YANG
Business Development
Manager



FAHMI HELMI
Business Development
Manager